ESPI



CONVERGING CURRENTS

RECOMMENDATIONS TO ALIGN SPACE AND WATER POLICY AGENDAS IN EUROPE

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CONVERGING CURRENTS

EXECUTIVE SUMMARY

This study identifies practical pathways for a stronger alignment between space and water policy priorities. It aims at informing ongoing and future policy processes, including the initiatives conducted in the framework of the European Water Resilience Strategy and the UN Water Action Agenda.

The results are derived from an analysis of national freshwater policy priorities across 32 European countries, as well as a compilation and review of over 250 primarily European-led downstream projects leveraging space data and infrastructure for freshwater-related applications, complemented by insights from stakeholder consultations.

National water policy priorities across European countries concentrate around mitigation of droughts and inland flooding, surface and groundwater pollution prevention, preservation of freshwater ecosystems, urban wastewater management and water infrastructure resilience, especially water supply management.

Satellite solutions already focus on surface water availability and hydrometeorological use cases such as floods and droughts, typically associated with high levels of insurance coverage, but still need to expand to more operational and localised applications for higher impact, addressing urban water management, water efficiency and sustainable water supply management. The integration of space applications with other technologies is proving essential but still untapped: only 20% of projects combine space with either AI, Internet of Things or Drones, addressing use cases such as irrigation management, green water and water losses.

HIGH CONTRIBUTION OF SPACE ACROSS THESE USE CASES:
Drought monitoring & management, Inland flood risk management, Green water availability, Surface water availability monitoring & management

At the intersection of the space and water policy communities, dialogue occurs across three main pathways: 1) top-down, initiated by national ministries or transboundary river basin authorities seeking technical support from international organisations, which connect requests with relevant geospatial expertise and networks (including international water conventions); 2) bottom-up, through satellite-based service coordinators engaging local water and disaster management actors, building capacity for using space-derived data and creating opportunities for service co-design, for instance hydrological predictions; 3) on the ground, with local actors supporting space applications by providing access to field sites, as well as in-situ information for satellite calibration and data validation.

Overall, a contrasted policy-project alignment is observed, with the level of prioritisation of water policy objectives being unevenly addressed by satellite solutions, in number and size.

Key opportunities to expand space solutions emerge. In particular, the value of space data seems to be overlooked for ground-level operations in water supply and urban wastewater management, despite demonstrated potential (e.g., leak detection, water reservoir stability monitoring, detection of untreated sewage contaminating water resources). The focus of space solutions on floods, droughts and irrigation management instead suggests that the space sector remains largely focused on established use cases and "passive" (yet essential) monitoring, and has yet to transition towards more "proactive" solutions addressing less-explored operational needs.

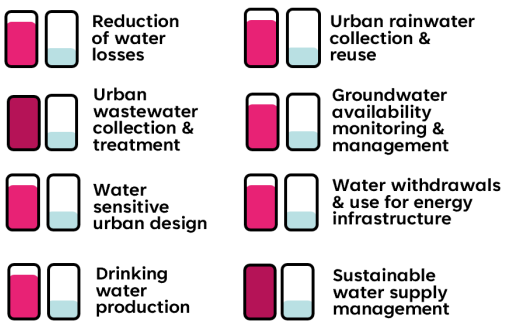
Water diplomacy has emerged as a high-level policy concept reflected in a significant number of water policies, some specifically dedicated to it. However, it has yet to be explicitly recognised as an objective that space-based solutions can actively support, both within space policies and as the target of satellite solutions. In many transboundary contexts where negotiations over shared water resources management are hindered by either the absence of data, outdated data, or disagreements over the credibility of datasets, stakeholders consistently emphasised the current and potential value of Earth observation as a trusted, third-party information source, helping to reduce information asymmetries and build trust between riparian states.

PROMISING AVENUE:
Cooperation on Transboundary Water Basins

OPPORTUNITY ANALYSIS

Comparison of the level of prioritisation of water policy objectives and their implementation through space-enabled projects (full graph is in §2.4 of the report):

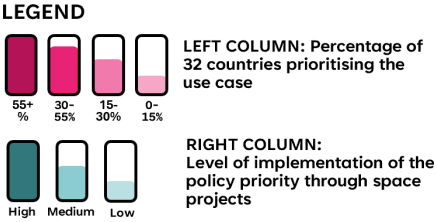
MOST PROMINENT GAPS



ESTABLISHED EFFORTS WITH FURTHER POTENTIAL



LEGEND



FINDINGS

Awareness of space applications within the water sector is high, but water-space policy alignment remains limited. While nearly 60% of analysed national water policies reference space technologies, mentions of water in space policies remain, in comparison, limited. Awareness gaps remain, as 20% of water policies mention only non-space technologies such as AI or drones. Overall, there is limited correlation between policy cross-reference and space projects implementation, which suggests that water and space policies evolve largely in parallel with limited coordination: water policies emphasise technological applications for water resource management (e.g., urban wastewater), while space policies tend to prioritise established use cases where the value of space is already demonstrated (e.g., droughts and inland flooding), rather than less-explored application areas (e.g., drinking water production, water supply management).

Water sector actors, public and private, are involved in only slightly over 25% of the consortia of analysed satellite-based projects. Public water actors (e.g., municipalities, public agencies), despite representing nearly half of primary end-users, are mostly absent from project consortia, creating a risk of technology providers disconnecting from implementation needs. Private water actors (e.g., water supply companies) remain also largely absent: when involved, they help connecting solutions with a broader spectrum of end-users, including hydropower producers, water supply companies and small-holder farmers.

Water management governance is decentralised, with implementation often occurring at basin and municipal levels. However, nearly 60% of analysed space projects are deployed at continent or global scale. While global, open-access platforms provide shared tools and datasets, local actors need tailored applications reflecting specific governance arrangements (e.g., river basin authorities) and water management challenges (e.g., wetland health monitoring). Limited funding for stakeholder mapping during project design reduces understanding of institutional landscape and implementation needs.

River basin organisations remain underrepresented in the consortium of space projects despite their strategic role in water management. Uniquely positioned to support national and transboundary water management, they bring valuable governance experience, GIS/HIS datasets, and long-term institutional knowledge. Integration of satellite data with these organisations' ground observations would strengthen the work of river basin organisations beyond flood and drought forecasting, for freshwater ecosystem management and water quality; this requires capacity building, peer learning, and resource support to address differing basin challenges.

Recent water policies (2019-2025) increasingly emphasise water security, encompassing water availability, water infrastructure resilience and water diplomacy, yet these priorities remain insufficiently addressed by space projects. Despite its policy prominence, water infrastructure resilience features in a few projects, leaving opportunities in water supply management, water-sensitive urban design and drinking water production. Stakeholders noted that space data's value is often overlooked in these activities traditionally viewed as ground-level operations, despite proven applications (e.g., reservoir monitoring, predictive maintenance of pipe networks). Similarly, while nearly 60% of space projects have an international impact reflecting the transboundary nature of water resources, water diplomacy remains largely absent from space policies and projects, despite being prioritised by nearly 30% of countries and the demonstrated value of space for transboundary cooperation.

Private companies offer untapped opportunities to finance and scale up water-related space solutions. Private water actors (e.g., private water suppliers, water management consultancies), involved in only 20% of projects analysed with little overlap with public water actors, provide access to end-users who can help design and finance solutions (e.g., insurers, large producers). Water-dependent companies also have incentives to reduce their environmental footprint and support river basin preservation and restoration, as their degradation can be a consequence of their activities and a direct threat to their continuity. Private companies sometimes also sponsor publicly developed, open-access space-based platforms, stimulating downstream demand.

Space data delivers the greatest value when combined with in-situ data and embedded in broader governance frameworks. Stakeholders cautioned that adopting an exclusively technology-driven approach to freshwater management without recognising where ground data becomes necessary would risk "overselling" the potential of contribution of space technologies to freshwater applications, potentially leading to misdirected investments. Additionally, they underscored how such a technology-driven approach may trigger rebound effects: in agriculture, for instance, tools enabling more efficient water use often lead to the expansion of the cultivated area instead of reducing overall water consumption, especially when policy measures to reduce consumption remain limited.

RECOMMENDATIONS

1 POSITIONING SPACE AS A CORE COMPONENT OF WATER MANAGEMENT SOLUTIONS

To encourage the development of solutions that directly address existing water policy challenges and improve the translation of policy priorities into space projects, there is a need to align cross-references between space and water policies around shared priority areas, where relevant. National space policies should explicitly recognise water as a strategic application area, while in countries promoting technology-enabled water solutions without reference to space, strengthening space considerations in water policies can help create entry points for the development of satellite-based solutions aligned by design with sectoral priorities.

2 CO-DESIGNING SPACE SOLUTIONS WITH WATER SECTOR ACTORS

Public and private water actors should be systematically involved as co-designers of water-related space projects rather than only consultees. Involving end-users throughout project development would strengthen trust, operational relevance and buy-in of solutions.

3 CONNECTING SPACE SOLUTIONS WITH WATER GOVERNANCE FRAMEWORKS THROUGH LOCAL ACTORS

Space solutions should be designed around the decentralised nature of water governance frameworks, informed by funded and mandatory stakeholder mapping. To enable this process, local and regional water actors should act as intermediaries to translate end-user needs into tailored solutions.

4 COORDINATING ACTION AT THE RIVER BASIN LEVEL

River basin organisations should be positioned as co-designers of space projects for water management, leveraging existing networks and peer-learning initiatives. Sustained dialogue with space providers should focus on needs identification and capacity building, moving from ad hoc collaboration towards shaping a shared agenda, to integrate space technologies into river basin management.

5 BRIDGING THE WATER-SECURITY-DIPLOMACY AWARENESS & OPERATIONAL GAP

The space sector should increasingly develop solutions addressing the operational needs of the water sector, particularly the security and resilience of water infrastructure. This should include stronger engagement at municipal level, where water supply is primarily managed, in anticipation of the growing prominence of water security in future policies. In parallel, the role of space technologies in diplomatic efforts and cooperation over shared water resources should be more explicitly recognised and translated into concrete objectives, both in water and space policies, and at implementation level.

6 LEVERAGING PRIVATE ACTORS AND FINANCE

Opportunities to co-finance water-related space solutions with private companies, and to increase the participation of private water actors in space projects consortia, should be actively explored. Such partnerships would help broaden the reach of space solutions and scale up applications addressing operational needs and preventive action (e.g., leak detection, water supply management).

7 RECOGNISING THE LIMITS OF SPACE TECHNOLOGIES AND MAXIMISING THEIR ADDED VALUE

Space-based services should complement, rather than substitute, in-situ data and local expertise. Integrating space and ground-based data will support adoption, while pairing investments with policy reforms on water allocation, pricing and abstraction limits will ensure that efficiency gains genuinely improve freshwater management.

TABLE OF CONTENTS

1 WATER HAS NO ENEMY: CONNECTING DATA AND RESILIENCE 6

1.1 Objectives and content of the study 7

1.2 Scope and structure of the study 7

1.3 Methodology 8

2 THE SPACE-WATER POLICY NEXUS 10

2.1 National Water Policy Priorities 10

2.2 Space and Water Policy Integration 13

2.2.1 Space and Non-Space Technologies Integration in Water Policies 13

2.2.2 Water Policy Priorities Integration in National Space Policies 15

2.2.3 Alignment between Water and Space Policy Priorities 15

2.3 Landscape of Existing Water-Related Space Projects 17

2.3.1 Main Space Projects Characteristics 17

2.3.2 Current and Emerging Data Sources and Technologies 18

2.3.3 Overview of Space Projects across Water Use Cases 19

2.4 Alignment Between Policy Priorities and Space-Based Water Solutions . 21

3 CONVERGING CURRENTS: STRATEGIC PATTERNS AT THE SPACE-WATER INTERFACE 26

3.1 Connecting Water Governance and Space Solutions 27

3.2 Private Water Companies: Complementary but Distinct Partners 30

3.3 Expanding the Financing Base for Space and Water 31

3.3.1 International Financial Institutions (IFIs) 31

3.3.2 Private companies 31

3.4 Space as an Enabler of Transboundary Water Cooperation 33

4 RECOMMENDATIONS 35

4.1 Positioning Space as a Core Component of Water Management Solutions 35

4.2 Co-designing Space Solutions with Water Sector Actors 35
4.3 Connecting Space Solutions with Water Governance Frameworks through Local Actors 36
4.4 Coordinating Action at the River Basin Level 37
4.5 Bridging the Water-Security-Diplomacy Awareness and Operational Gap 37
4.6 Leveraging Private Actors and Finance 38
4.7 Recognising the Limits of Space Technologies and Maximising their Added Value 39
4.8 The way forward 40

ACKNOWLEDGMENT 42
AUTHORS 43
ANNEXES 44
Annex A – Use cases description 44
Annex B – Detailed Methodology 48

1 WATER HAS NO ENEMY: CONNECTING DATA AND RESILIENCE

Access to water for domestic and personal use is a fundamental human right,[1] yet water resources face growing pressure from scarcity and increasing extreme events affecting many sectors, including impacts on energy production.[2] In 2026, the concept of "Global Water Bankruptcy", further highlighted the depletion of both renewable water "income" from rivers and soils, and long-term "savings" held in aquifers, glaciers, and wetlands.[3] Meanwhile, previous ESPI research identified freshwater use cases such as water quality as high priorities within climate resilience and agriculture policies.[4]

As the Nigerian songwriter Fela Kuti sang, *water has no enemy*,[5] especially when observed from the bird's-eye-view perspective of satellites. Water is an essential shared resource that transcends individual nations or sectors. Investing in water means investing in health, biodiversity, the energy transition, climate resilience, food security, peace and diplomacy.

More than ever, effectively addressing water challenges depends on the availability and integration of timely and spatially comprehensive data. Space technologies – Earth observation (EO),[6] Positioning, Navigation & Timing (PNT) and Satellite communications (Satcom) – already support European policy priorities across domains, including freshwater management.[7] They enable the monitoring, forecasting, and transboundary cooperation required for sustainable water governance. Recent missions like ESA's HydroGNSS measure key hydrological climate variables, including soil moisture, wetland inundation and freeze-thaw states. Upcoming space missions like LSTM, CHIME and ROSE-L will further enhance the monitoring of evapotranspiration, land and surface water temperature, lake ecosystems, and the cryosphere.

This study presents a deep dive into space solutions for the freshwater domain, building on the findings of ESPI's report *Where Sensing Makes Sense — Opportunities for Advancing Europe's Priorities through Earth Observation,*[8] developed in collaboration with the European Space Agency (ESA). The report identified water as one of the priority policy domains with strong potential for collaboration with the space sector and highlighted the role of space as an enabler of water policy objectives.

[1] UNGA, Resolution 64/292 – The human right to water and sanitation, 2010. (Link)
[2] In 2026, low river water levels revealed the vulnerabilities of Europe's energy industry. See notably K. Than, L. Ilie, "Record-low Danube exposes climate risk to Hungary's and Romania's nuclear power", Reuters, 13 August 2026. (Link)
[3] UNU-INWEH, Global Water Bankruptcy, 2026. (Link)
[4] ESPI, Where Sensing Makes Sense, 2026. (Link)
[5] Fela Kuti, "Water No Get Enemy", 1975. (Link)
[6] World Bank Group, Earth observation for water resources management: current use and future opportunities for the water sector, 2016. (Link)
[7] ESPI, A Policy Vision for Earth Observation, 2025. (Link)
[8] ESPI, Where Sensing Makes Sense, 2026. (Link)

1.1 Objectives and content of the study

The study identifies practical pathways for **space and water policymakers to better align their agendas and accelerate the development and uptake of space-enabled solutions for water policy objectives**. While ESPI's previous research analysed how Earth Observation could address European policy priorities across domains (e.g., energy, agriculture, climate resilience), the present study focuses on a single sector: water, whose specific standards and operational logic call for a dedicated assessment. To this end, the study provides:

- An analysis of national freshwater policy priorities;
- A compilation and review of space-enabled freshwater projects and their alignment with policy priorities, including the identification of gaps and opportunities;
- A complementary assessment of the integration of space and non-space technologies into water policies, and of water considerations into space policies;
- Insights from stakeholders which, combined with other collected data, inform the identification of strategic patterns and opportunities for space and water collaboration, as well as actionable recommendations.

1.2 Scope and structure of the study

The study considers **blue water** (found in rivers, lakes, groundwater, aquifers) and **green water** (held in soil moisture, taken up by plants), accounting for both visible and often-overlooked components of the water cycle.[9] It covers all **space applications contributing to water monitoring**: Meteorology, Earth observation (EO), Positioning, Navigation & Timing (PNT), Satellite communications (Satcom).

The analysis spans **32 countries**, selected based on their membership in – or affiliation with – the **European Space Agency (ESA)**, the **European Union (EU)**, and/or the **European Economic Area (EEA)**.[10] This broader framework extends beyond the EU policy ecosystem while encompassing all European countries with a space policy (see Annex B).

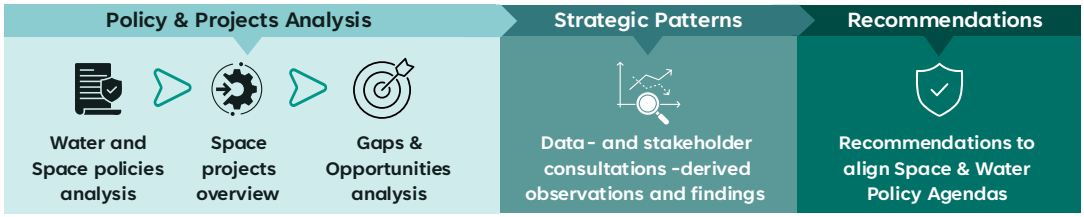


Figure 1 – Structure of the Report

[9] Definitions of blue and green water were developed by the International Water Management Institute (IWMI). (Link)
[10] On the future outcome of a global outlook on the topic, see §4.8 .

1.3 Methodology

Consistent with the study's objectives, the analysis required a screening of sectorial policies and projects, primarily conducted through desk research combined with data science tools, and complemented by a stakeholder consultation process consisting of multiple expert interviews to help shape and validate preliminary findings, identify patterns and mapping preliminary results against real life practice (See Figure 2).

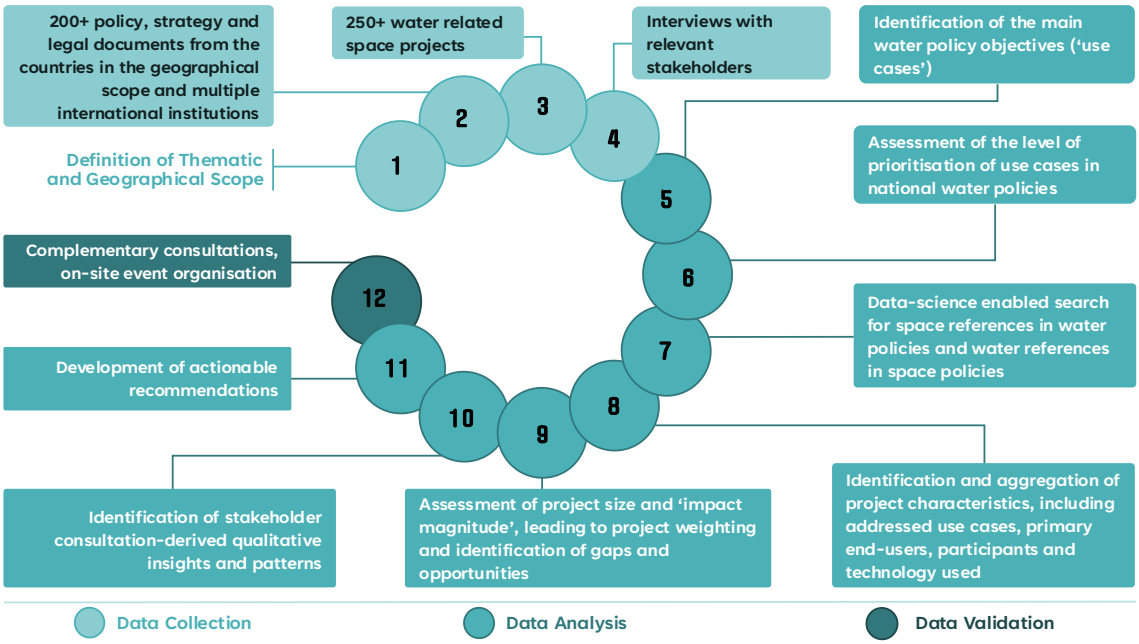


Figure 2 – Overview of methodology steps

30 freshwater-related policy objectives ('use cases') were identified at the national level, relying on policy and strategy documents with a clear and general focus on freshwater, excluding those focusing on one single use case (e.g., heavy precipitations) and those older than 2016 (unless being the only available national water policy). Using data science tools, a complementary assessment was conducted on the extent to which space and non-space technologies are integrated in water policies, as well as how water is integrated in space policies.

In terms of space applications, **more than 250 ongoing and planned projects* addressing freshwater use cases using space technologies** were selected (EO, PNT and Satcom), including those integrated with non-space technologies (e.g., drones). Details on the project selection methodology and their national distribution can be found in Annex B. It should be noted that the comprehensiveness of project collection is limited by the accessibility of project information online, and that the definition of "project" adopted in this report focuses on downstream solutions.

***Project – A funded initiative focused on the research, development, testing, deployment, or scaling of space-based solutions and services for a specific objective. It encompasses four different output types:**

Programme – A wider initiative involving several operational activities and (often longer-term) objectives, extending beyond single technical outputs.

System/Platform – Final output is a deployed, operational system or application (often online) translating space data into a specific service, tool or product.

Pilot Initiative – Research projects that have moved beyond the theoretical phase and are actively validating/testing solutions or services in a simulated or real-world environment, including operational pilots.

Research/Feasibility – Time-limited initiatives focused on early-stage research, with the primary output being theoretical knowledge, a feasibility study, or a research paper.

Building on a new methodological approach compared to previous ESPI research, each project has been assigned an 'impact magnitude score' to determine a project's weight by measuring its size and capacity to be sustainably funded with long-lasting impact (See Annex B). This assessment was used to compare the main objectives (i.e. use cases addressed) of the project to the level of policy prioritisation of the same use cases in national water policies, leading to the identification of gaps and potential opportunities. This accounts for the fact that some apparent 'gaps' in comparing the level of policy prioritisation and number/size of projects reflect limited applicability of space-based solutions rather than unmet potential for additional benefits. The analysis of the technical feasibility of space solutions which could fill existing implementation gaps is beyond the scope of this study.

Finally, strategic patterns were shaped in a sense-making exercise by combining data-derived observations from the policy analysis with insights from collected space projects and the stakeholder consultation. They provide a more comprehensive basis for actionable recommendations for space and water policy alignment.

Insights from the ESPI Vienna Space Diplomats (VSD) Evening Event on Space for Water Diplomacy[11] have also significantly contributed to shaping and validating initial results. The detailed methodology can be found in Annex B to this report.

[11] Vienna Space Diplomats (VSD) evening event, "Delivering water diplomacy through space", 15 June 2026. (Link) ESPI co-organised an evening event with the Permanent Mission of Slovenia in Vienna, focused on the role of space-based capabilities in strengthening water management, security, and diplomacy, and on identifying concrete actions for future cooperation. The discussion contributed to a broader Space for Water policy agenda, drawing on perspectives from diplomacy, Earth observation, and operational space applications. The high-level speaker line-up figures in the Acknowledgments of this report for their contribution to shaping and validating initial results.

2 THE SPACE-WATER POLICY NEXUS

Increasing alignment between water and space policy agendas begins with establishing an overarching understanding of the needs identified by countries and whether space plays a role in their policy and operational responses. This section includes an analysis of which freshwater priorities are reflected in water and space national policies (§2.1) and the extent to which they are integrated in space applications (§2.2). Comparing policy priorities with the current landscape of water-related space projects enables the identification of areas of alignment, opportunities and remaining gaps (§2.3).

2.1 National Water Policy Priorities

The use cases were identified through a qualitative screening of sectorial policies, strategies and regulations with a clear relevance to freshwater – water policies, but also climate resilience and agriculture policies – at international, European and national levels, published primarily between 2016 and 2025 (older policies were excluded, unless being the only available national water policy - see Annex B for the full use case description).

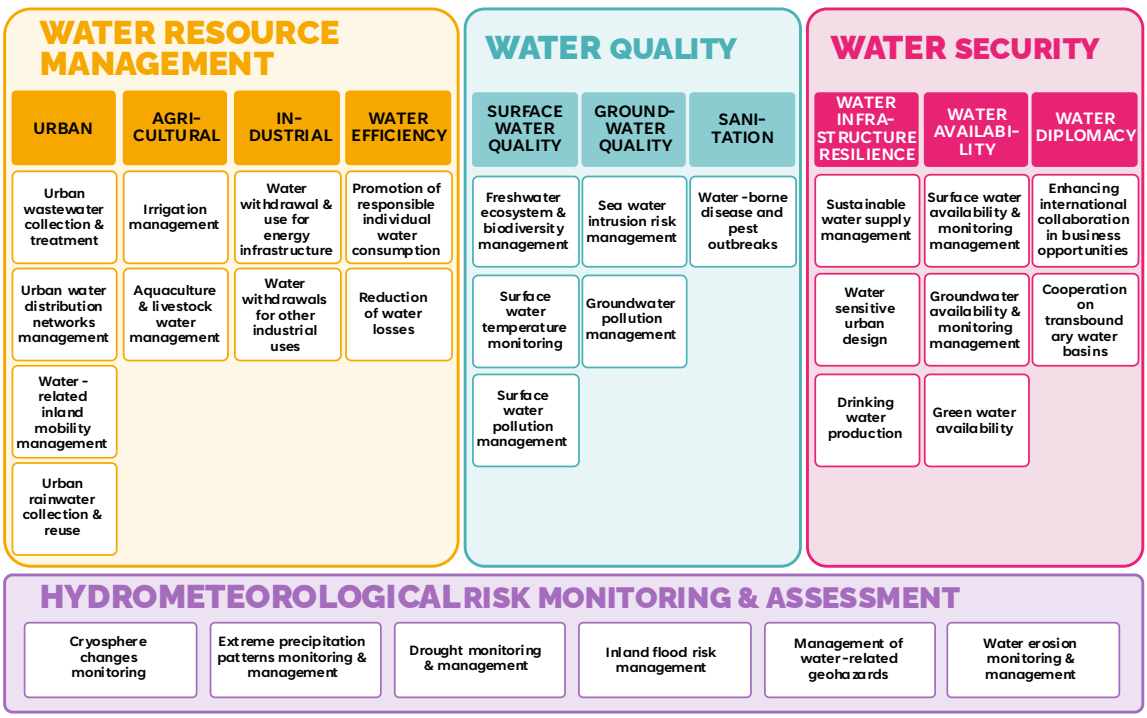


Figure 3 – Identified use cases, organised in broader clusters

Nearly 50% of the use cases presented above are security-related, encompassing both the security and resilience of water infrastructure and resources (*Water security* cluster) and the security risks arising from water-related events (*Hydrometeorological risks* cluster), which have cascading impacts across all other use cases.

Five use cases emerged as policy priorities in the vast majority of national water policies: (1) *Urban Wastewater Collection & Treatment* (72%), (2) *Surface Water Pollution Management* (66%), (3) *Inland Flood Risk Management* (59%), (4) *Freshwater Ecosystem and Biodiversity Management* (59%), (5) *Sustainable Water Supply Management* (56%).

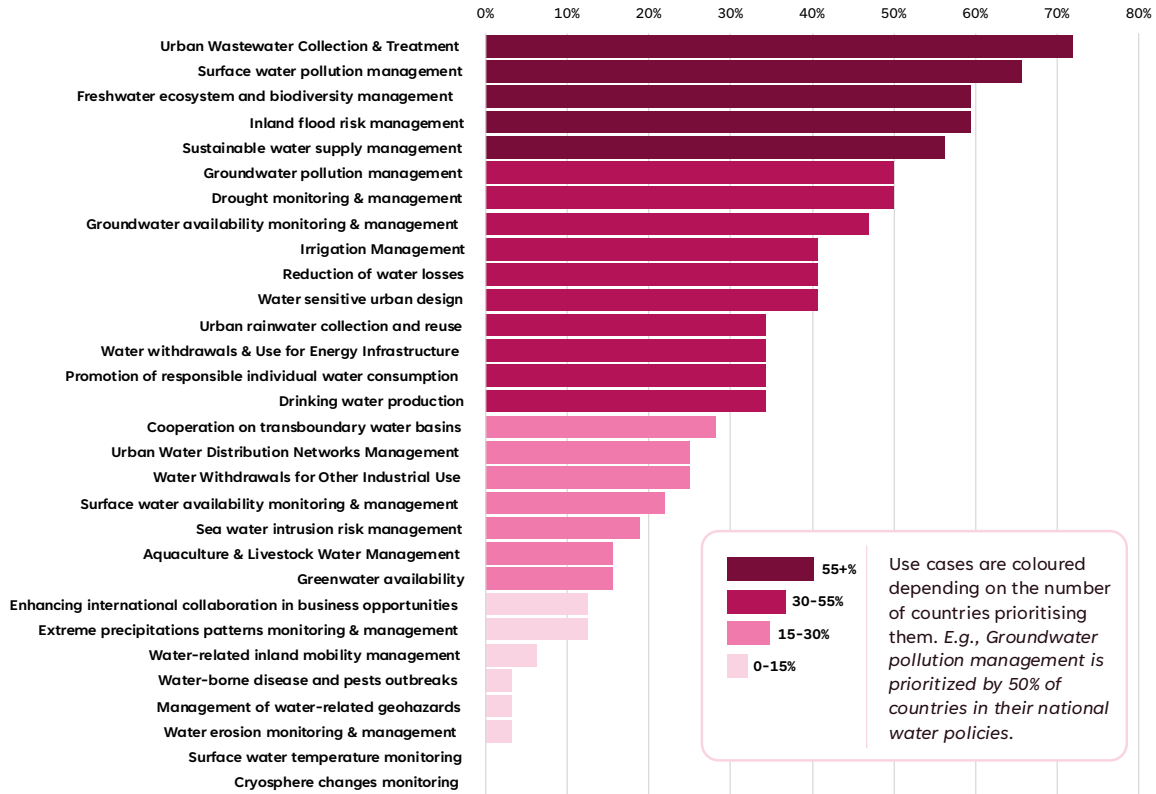


Figure 4 - Use cases prioritised in national water policies, organised from largest to smallest share of countries prioritising the use case.

The use cases are organised into clusters to facilitate analysis and readability and should not be interpreted as scientific classifications (See Figure 3). Rather, **each use case represents a policy objective** that can translate into one or more applications. Use cases within *Hydrometeorological risk monitoring and assessment* cluster should be understood as underlying drivers influencing many other use cases.

Use cases were then classified as **national policy priorities** when national water policies identified concrete measures to address them. (See Annex B)

Among the identified national water policy priorities, detailed in Figure 5:

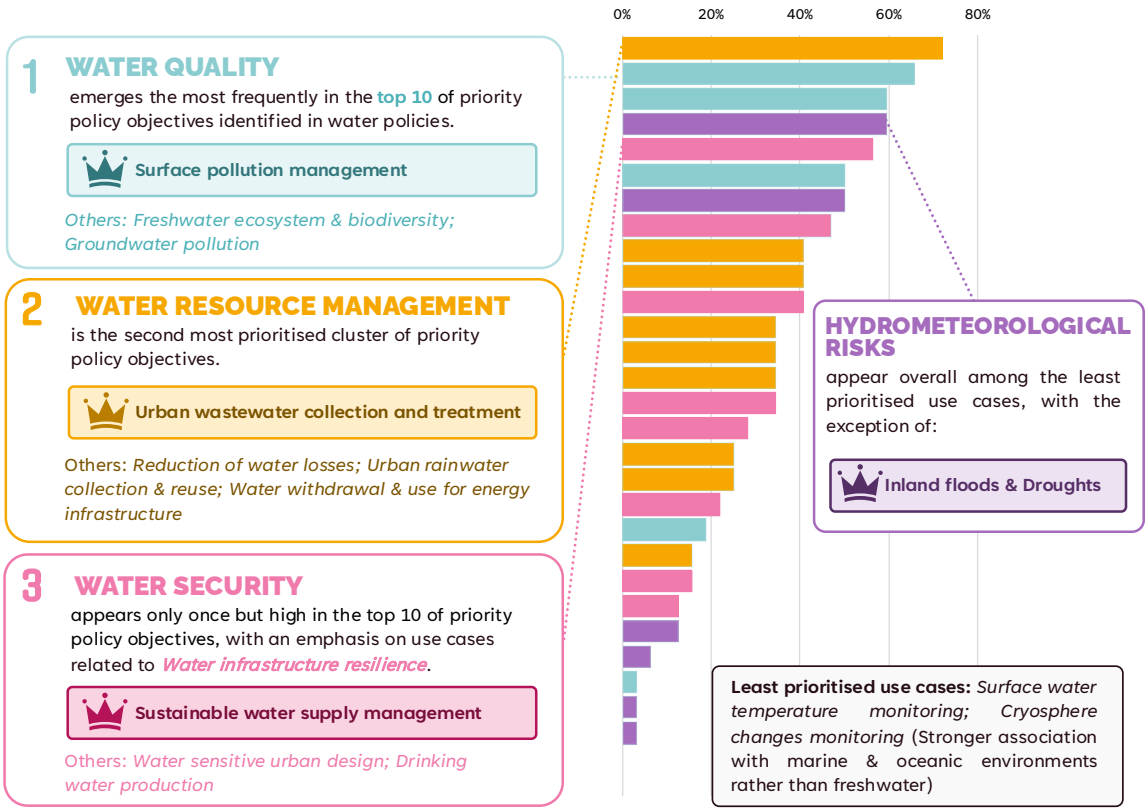


Figure 5 - National water policy priorities analysis

National trends

Generally, *water security* use cases receive greater emphasis in recent policies (2019-2025), suggesting the growing recognition of the security dimension of water resources. This cluster remains less prioritised overall, potentially because water-related conflicts occur predominantly outside Europe. For instance, **Germany** (2023) puts a great emphasis on all use cases linked to *water security*, including the continuous availability of water resources and cooperation over shared waters. Insular countries like **Cyprus** (2019) and **Malta** (2024) also largely prioritise *water infrastructure resilience* with an emphasis on *drinking water production*. **Croatia** (2009), on the other hand, prioritises *water resource management*, with a particular emphasis on *water efficiency.*

2.2 Space and Water Policy Integration

Integrating water and space policy agendas requires a two-fold approach: (1) understanding whether space applications are recognised as tools for addressing national water policy priorities, and (2) whether water is framed as an application domain in national space agendas. References to non-space technologies, such as IoT, in water policies also indicates a broader appetite for data-driven solutions beyond space applications. This section examines the integration of space and non-space technologies in national water policies, of water considerations in national space policies, and compares the results.

2.2.1 Space and Non-Space Technologies Integration in Water Policies

Freshwater policies demonstrate significant integration of space and non-space technologies, with a quarter of all countries referencing both. Nearly **60%** of screened countries made references to space technologies and applications in their freshwater-related policy documents, while around **45%** referenced non-space technologies with references distributed evenly across Europe. (See Figure 5)

Around 20% of countries refer to **neither space nor non-space technologies** in the screened policies. These include France and Italy, despite managing the largest share of space projects together with Germany and the UK (see §2.3.1).

In some countries, results show a significant policy appetite for solutions involving non-space technologies, but a lack of awareness of existing or potential space solutions. In fact, nearly 20% of countries reference non-space technologies – broadly (e.g., "emerging technologies"), or more specifically (e.g., "IoT") – without mentioning space. This shows a potential awareness gap for certain use cases, such as *drinking water production*.

Urban wastewater collection and treatment appears the most (nearly 20%) in non-space technologies references, followed by *reduction of water losses*, *surface water pollution management* and *inland flood risk management*. (See all use cases in Figure 14, Annex B).

National trends

Spain has the highest number of mentions of both space and non-space technologies, followed by **Cyprus** and **Austria** for space mentions. National water policies vary considerably in length, but this alone does not explain differences in (space) technologies integration. Longer documents may allow for more technological references through greater operational detail (e.g., Spain, Austria), yet several lengthy policies contain few or none (e.g., France, Italy, Sweden, Liechtenstein). Conversely, **Cyprus** and **Slovenia** stand out for their numerous space mentions despite relatively short documents.

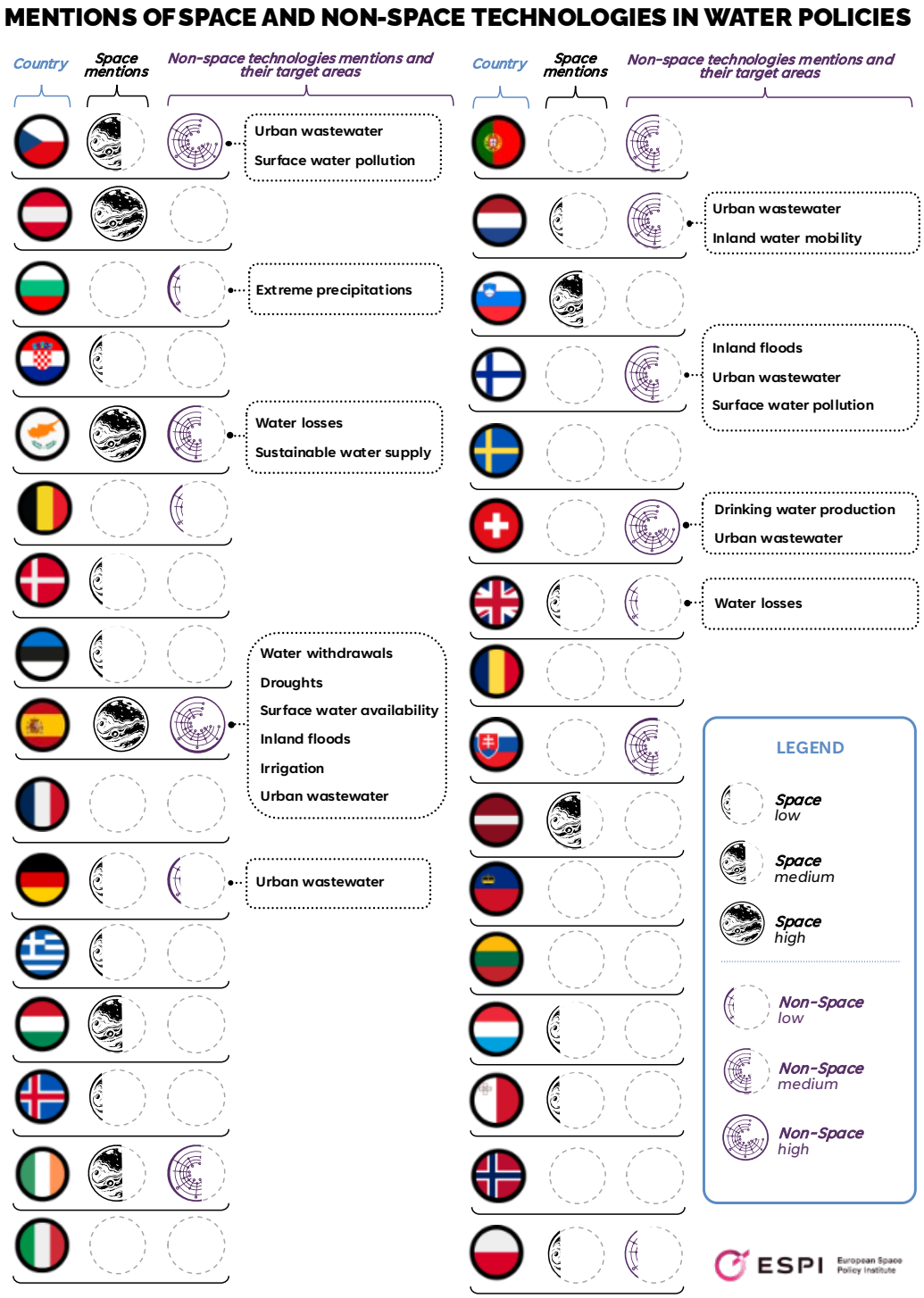


Figure 6 - Mentions of space and non-space technologies in national water policies, with priority areas targeted through non-space technologies (when applicable)

2.2.2 Water Policy Priorities Integration in National Space Policies

Regarding the other side of the policy agenda, there is room for further integration of water considerations into national space policies (See Figure 6). Slightly over **50%** of all countries reference water in their space policies, with **35%** making only a single reference to water. Across space policies, nearly 80% of water mentions focus on a specific use case: *Inland flood risk management* appears in more than 20% of water mentions – showing **a tendency for space policies to prioritise responses to challenges within established use cases**, where the value of space is already demonstrated, rather than addressing less-explored issues and emerging application areas. It is followed by *Drought monitoring & management*, and *Irrigation management* and *Surface water availability monitoring & management,* cited in 12% of mentions (See Figure 15).

National trends

Cyprus and the **Netherlands** count the highest number of references to water in their space policies, followed by **Germany**. Use cases mentioned by each country in their water and space policies largely differ, with the exception of **Cyprus** and **Finland**, which address the same use case in both policies; the former with *Sustainable water supply management*, the latter with *Inland flood risk management*.

2.2.3 Alignment between Water and Space Policy Priorities

A comparative analysis of water and space policies reveals where priorities align or diverge. The freshwater use cases emphasised in water and space policies mainly differ, suggesting that **policy interest in technology-enabled solutions is often misaligned with the awareness of where space technologies can contribute**.

The clusters most addressed by space policies are *Hydrometeorological Risk Monitoring and Management* and *Water Security*, while water policies mainly mention general technology solutions in relation to *Water resource management* (See Table 1). It is, however, essential to place this misalignment in the broader context, considering the fact that not all countries with a water policy also possess a national space policy.

Freshwater use case (clusters)	Mentioned alongside non-space technologies in water policies	Mentioned in space policies
Water resource management	55%	20%
Water quality	10%	10%
Water security	15%	30%
Hydrometeorological Risk Monitoring and Management	20%	40%

Table 1 – Percentage of use cases mentioned in water and space policies per cluster

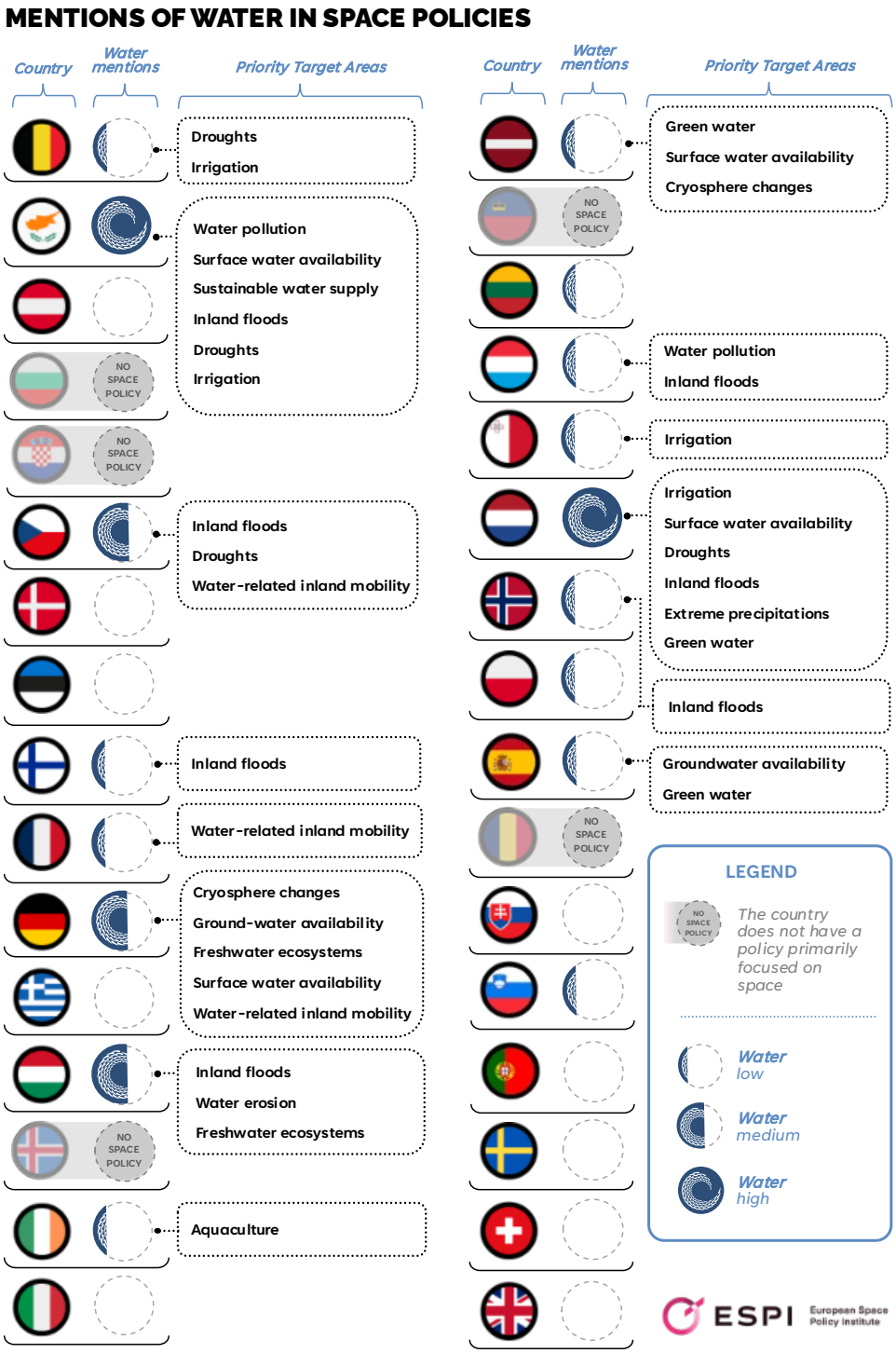


Figure 7 - Mentions of water in national space policies, with identified target areas (when applicable)

2.3 Landscape of Existing Water-Related Space Projects

While policy integration encourages the adoption of space-based solutions and the allocation of resources, it does not guarantee practical implementation. It is therefore crucial to examine the extent to which national priorities translate into operational space-based projects. This section provides an overview of the primary statistics of the projects collected, followed by a deep dive into their technical foundations and the water policy priorities they address.

2.3.1 Main Space Projects Characteristics

The analysis of 250+ space projects across 32 countries reveals that space-based applications for freshwater priorities rely overwhelmingly on Earth observation data (95%). While most operate at a large scale, a significant number of projects are implemented at the local level, reflecting the site-specific nature of water management. Furthermore, achieving maximum impact often requires combining space data with complementary inputs, with **only one in five projects reliant on the integration of non-space technologies**, particularly AI and Machine Learning, to translate space data into operational services. An overview of the projects collected is shown in Figure 8 below, providing the main statistics on project type, output, scale, managing region, and primary data sources. (See the definitions of each element in Annex B).

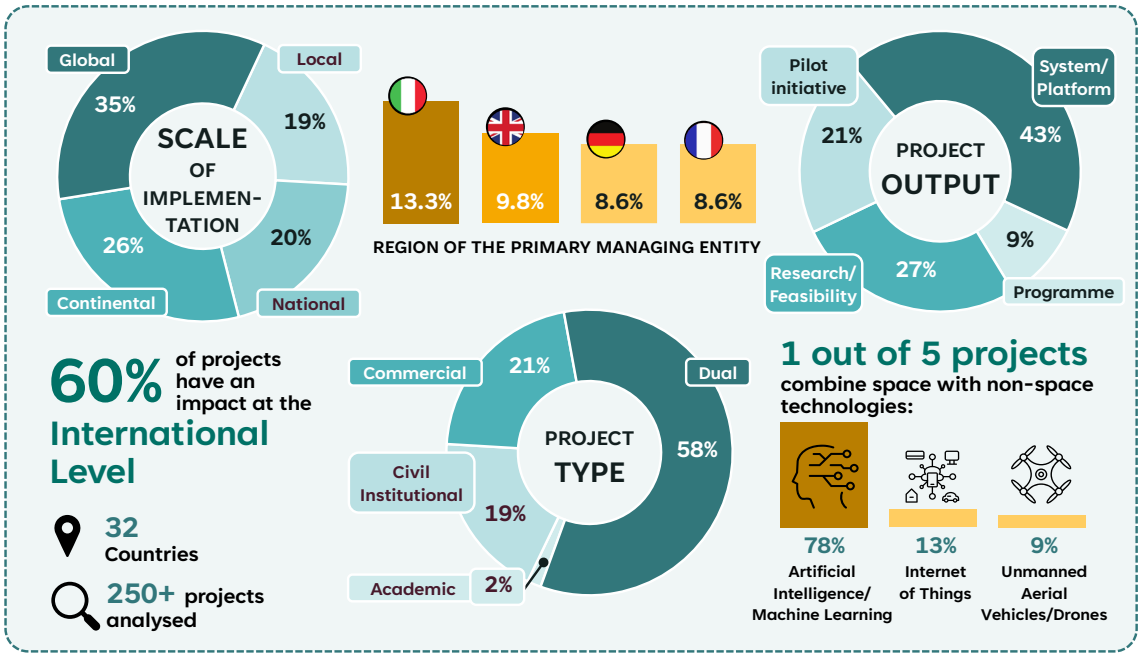


Figure 8 – Overview of the main space projects characteristics

Notably, a disconnect can be observed between national policy integration and project implementation in the space-water sector. Italy, France, Germany, and the UK manage the largest share of water-related space projects, despite having very limited cross-references between space and water in their national policies (See §2.2.1 & §2.2.2). Conversely, policy focus does not guarantee implementation. For instance, Cyprus heavily cross-references both sectors in its national policies, ranking alongside the Netherlands, Germany, Spain, and Austria, yet manages very few active projects, whether through the participation of national authorities and ministries in consortia or through companies headquartered in the country. While such gaps can also emerge due to differences in national space capabilities and industrial policy frameworks, this highlights the need for increased alignment between the integration of space and water in policies with concrete project implementation.

2.3.2 Current and Emerging Data Sources and Technologies

In terms of space technologies used, 94% of analysed space projects are enabled by Earth Observation, 4% combine EO with GNSS and 2% use solely GNSS, while a few also leverage meteorological data. Satellite communication primarily supports field data transmission from remote areas without internet connectivity (e.g., from remote water pipelines) and rarely appears as a project's main space technology used. GNSS-enabled projects, while in minority, draw on specific sensing methods for geohazards monitoring, such as GNSS Reflectometry (GNSS-R) with applications focused on ocean, ice and land monitoring. It is the technology used for instance by ESA's recent mission HydroGNSS, which measures soil moisture, wetland inundation and freeze-thaw states, among other hydrological variables. Beyond the monitoring of geophysical parameters, GNSS-enabled projects also address operational use cases such as water leak detection and water quality monitoring.

Satellites from the **Sentinel** and **Landsat** constellations constitute a major share of the data sources used by space projects, alongside **Prisma** and historical data from **ERS-2** (European Remote-Sensing Satellite-2). The resulting applications primarily rely on data acquired through multispectral optical sensors, synthetic aperture radar (SAR), thermal infrared sensors, and microwave radiometers.

Sentinel-1	Sentinel-2	Sentinel-3	Sentinel-5P	Landsat-5
Landsat-6	Landsat-7	Landsat-8	Landsat-9	ERS
Prisma	EnMAP	Cosmo-SkyMed	SAOCOM	ANSER
Φsat-2	CryoSat	SMOS	ENVISAT	AMSR
TanDEM	GRACE	GPM	AQUA	TERRA
METOP	Ecostress	Iride	SMAP	ASCAT
WorldView	IceSat	BIOMASS	ROSE-L (planned)	LSTM (planned)
CHIME (planned)	Galileo	HydroGNSS	CYGNSS	

Table 2 – Data sources used by space projects (Most used data sources appear in pink)

Certain projects plan on using data from **future missions**, such as LSTM (15-meter ground resolution for Evapotranspiration, land and water surface temperature) and CHIME (30-meter ground resolution) for measuring soil moisture, with secondary applications of its

Hyperspectral Imager in areas such as water quality and lake ecosystems. The future ESA mission ROSE-L will also provide high-resolution soil moisture data below vegetation canopies and enhanced monitoring of the Arctic and cryosphere. [12]

Only one out of five collected space projects combine satellite data with other data sources and technologies, which mainly consist of Artificial Intelligence and Machine Learning (AI/ML), Internet of Things (IoT), and UAV/Drones. Among space projects using non-space technologies, AI/ML emerged as the most common (78%), compared to IoT (13%) and UAV/Drones (9%) (See Figure 8). Projects using IoT and drones generally address green water availability and irrigation management, and to a lesser extent water losses, droughts and inland flooding. Meanwhile, AI/ML-driven models are widely integrated with satellite data to address numerous use cases; stakeholder consultations underline in particular their role in floods and flash floods prediction, serving as a growing complementary technology to physically based models, both of which can use satellite data as input (for physical based models) or training data (for AI-driven models).

Stakeholder consultations revealed that some organisations prioritise ease of access over high precision when selecting the data powering their platforms. In the cases brought forward, the stated objective is to ensure that data complements, rather than replaces, local knowledge and equips local water management actors with the basic knowledge and tools to improve their capacities.

2.3.3 Overview of Space Projects across Water Use Cases

The space projects directly relevant to the freshwater domain (250+ identified) are distributed across the use cases shown in Figure 9. This distribution only includes projects explicitly mentioning one or more of these use cases within their objectives and does not account for any indirect or potential relevance of projects for other use cases.

Rather than the number of projects, Figure 9 displays the overall **level of implementation of water-related policy objectives through space** for each use case. As detailed in Annex B, each project was assigned a weighted score based on criteria such as budget, stakeholder involvement, and target end-users, prioritising projects with sustained funding and greater potential for long-term impact. The weighted scores were then summed and converted into a *low-medium-high* scale to define the current level of implementation of policy priorities through space projects. This method applies solely to use case assessment and is not relevant for other characteristics, such as “scale of implementation” or “project type”.

[12] ESA, LSTM, (Link); CHIME, (Link); and ROSE-L. (Link)

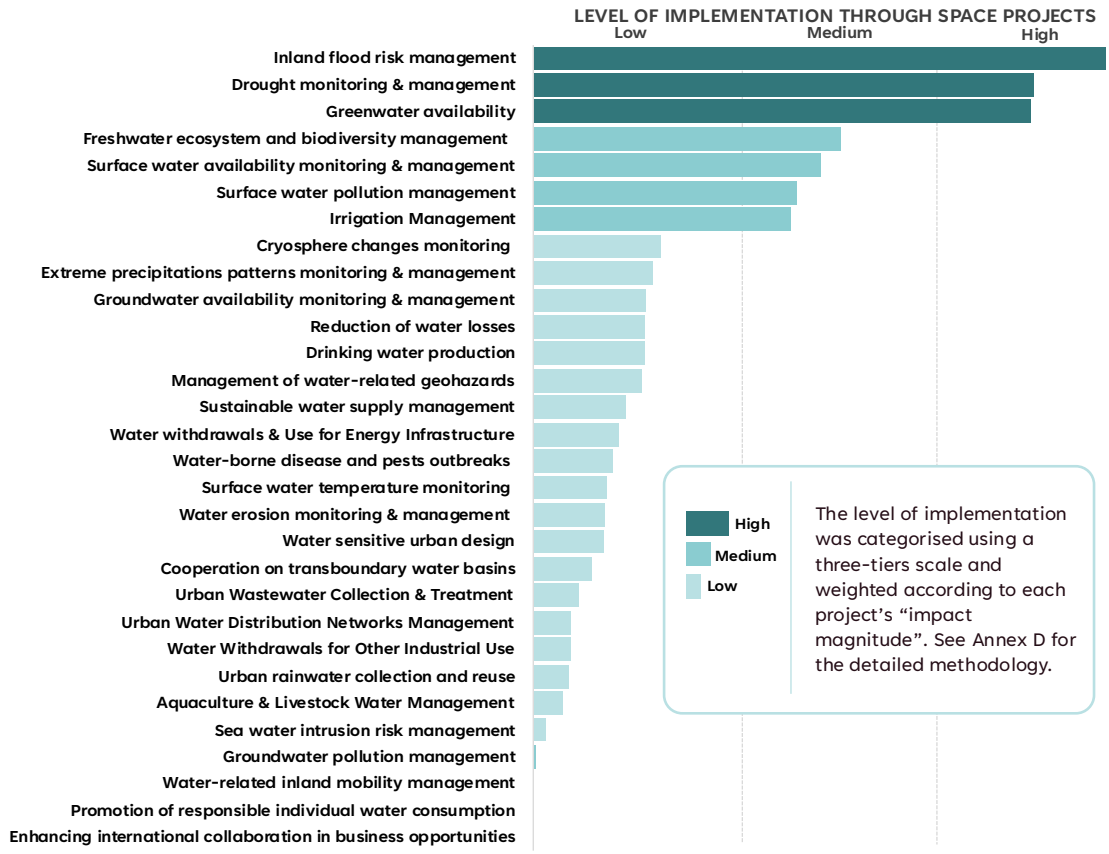


Figure 9 – Level of implementation of water-related policy objectives (use cases) through space projects.

Overall, projects are primarily concentrated in three core use case categories: (1) *hydrometeorological risks*, with a strong emphasis on *droughts and flooding*, (2) *water availability*, particularly regarding surface and green water, and (3) *surface water quality*. Outside of these three categories is the emphasis on *irrigation management*, primarily in relation to precision agriculture. While these remain high-priority use cases, the comparatively limited contribution of space projects to other key objectives underscores the **need for a more systematic and comprehensive deployment of space-based solutions** across the freshwater domain. In particular, there is significant potential to build on existing 'passive' observation and monitoring (e.g., for droughts and inland flooding) and move towards 'active' prevention and mitigation (e.g., leak detection in water reservoirs and pipelines, water-sensitive urban design).

Many projects address multiple use cases simultaneously, demonstrating the **versatility of space technology to support distinct policy objectives** with the same instruments and datasets. However, projects addressing multiple use cases remain largely concentrated

within traditional risk management and monitoring domains. For instance, a high degree of overlap is observed between flooding and drought monitoring, capturing the compound nature of hydrometeorological disasters where preceding drought conditions exacerbate runoff and subsequent flood risks. Similarly, strong co-occurrence is visible across *surface water availability, drought, and surface water pollution*, illustrating how space solutions are increasingly used to monitor the combined impacts of water quantity and quality degradation.

Significant overlap was also identified between *green water availability* and *irrigation management*, largely driven by the reliance on soil moisture data as a key indicator for soil health and precision agriculture. While the increasing focus on irrigation management demonstrates a shift beyond passive monitoring towards proactive resource management, these application-oriented solutions need to be scaled further to better align with the level of policy prioritisation and maximise operational impact. A similar operational link is observed in projects bridging the use cases on cryosphere monitoring and energy infrastructure, where remote sensing data is used to track snow cover and melt dynamics, supporting hydropower optimisation, plant planning, and daily operations.

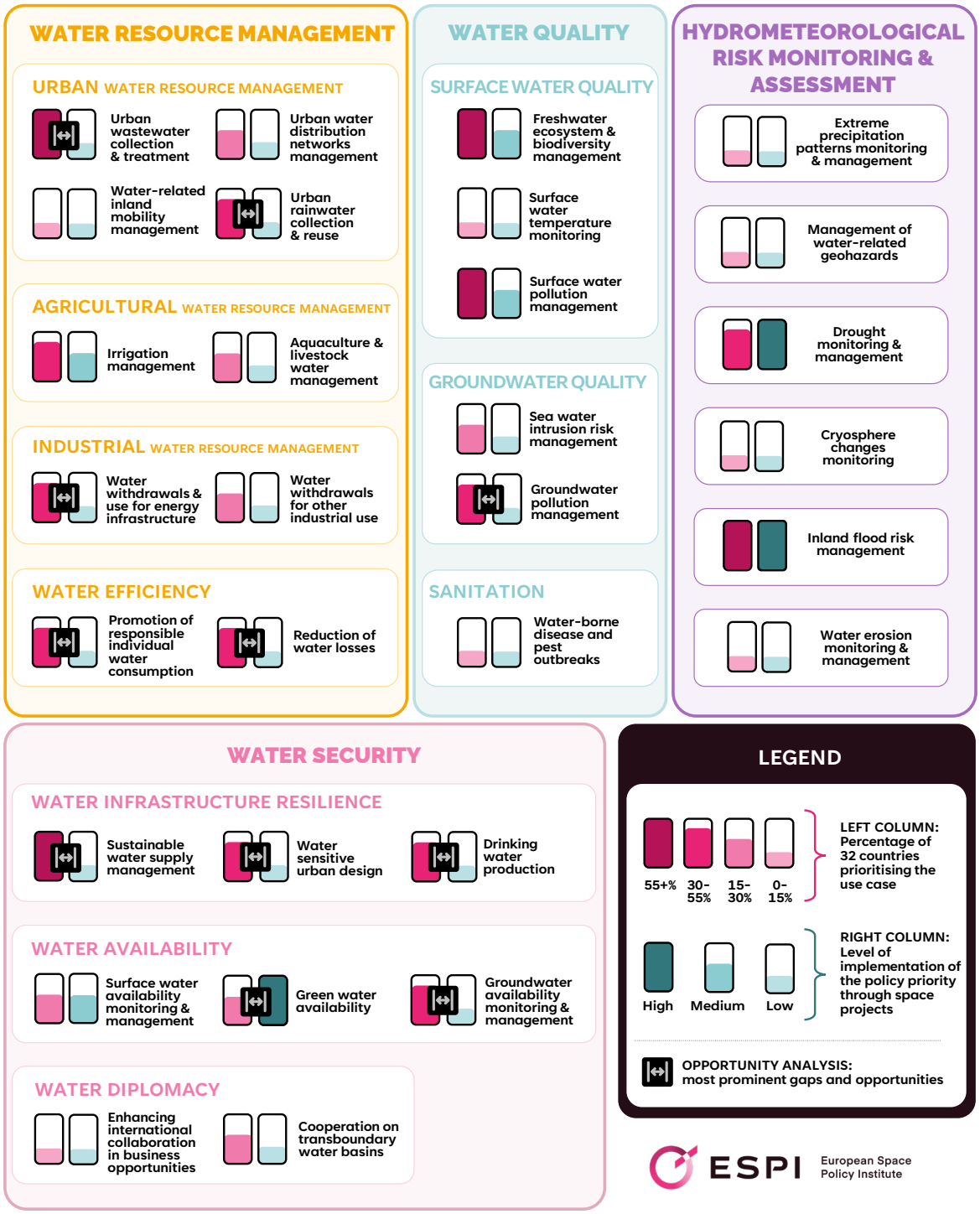
However, **minimal co-occurrence was identified in areas requiring more specific, localised solutions or integration with in-situ data, such as *urban water management and supply*.** Furthermore, the limited uptake of space projects addressing transboundary and international cooperation highlights a need to better leverage space-based solutions for water diplomacy. Water diplomacy has significant potential for cross-cutting applications that can address multiple freshwater priorities simultaneously, while also contributing to broader international relations and security objectives (See §3.4).

2.4 Alignment Between Policy Priorities and Space-Based Water Solutions

Comparing freshwater policy priorities against space-based projects reveals contrasted alignment,[13] with high-priority policy areas generally showing strong project uptake. Identifying discrepancies between policy emphasis and project implementation highlights key opportunities to expand space-based solutions. One important gap can be observed for *Water infrastructure resilience* use cases, including *sustainable water supply management*, *water-sensitive urban design*, and *drinking water production*. However, not all gaps indicate unmet potential, with some simply reflecting the technical limitations of space technologies for certain use cases – the analysis of the technical feasibility of space solutions which could fill existing implementation gaps is beyond the scope of this study.

[13] Although compared to other domains analysed in ESPI's previous research such as Energy, the Water domain presents a relatively high alignment between policy priorities and their level of implementation through space projects. (Link)

GAPS & OPPORTUNITIES



ESPI European Space Policy Institute

Overall, contrasted alignment is observed between policy objectives and active space-based projects. While there generally tends to be fewer related projects where there is less policy focus, use cases highly addressed by policies are not always represented by an equivalent level of implementation through space projects, taking into account the type, size, and impact magnitude of these projects.

Particularly strong alignment is visible in the *hydrometeorological risk monitoring & management* cluster. However, since these applications act more as underlying drivers than standalone use cases, they should not be evaluated in isolation. Their high prevalence reflects how deeply interlinked they are with other use cases (e.g. drought directly reduces water availability), while also being tied to the fact that these hydrometeorological risks – in particular floods and droughts – are commonly covered by insurance products, creating demand for technology-driven solutions that can monitor, predict and quantify their frequency, severity and characteristics.[14]

More **specific opportunities** can be identified in the following use cases:

- *Urban water management* – Use cases within this category show particularly low operational implementation. **Despite being prioritised by 72% of countries, *urban wastewater collection & treatment* remains under-addressed by space projects.** While expert consultations revealed that its operational and localised nature makes space applications less obvious, remote sensing can help to identify suitable sites for wastewater infrastructure while protecting surrounding water sources. There is similarly low alignment for *urban rainwater collection and reuse*, despite the capacity of space technologies to detect and combine vital indicators such as soil moisture, elevation, and slope.[15] **Adapting these applications to urban contexts by integrating space-based indicators with in-situ data would support municipalities in optimising both rainwater collection planning and reuse** (e.g. urban green space irrigation).
- *Water infrastructure resilience* – *Sustainable water supply management, water-sensitive urban design,* and *drinking water production* are high policy priorities that remain under-addressed by space projects. Stakeholder consultations highlighted that **since water supply and sanitation are considered ground-level operations, the value of space data is often overlooked.** However, existing projects demonstrate how space technologies are already being leveraged to optimise pumping schedules during low-flow periods, monitor reservoirs, and enable predictive maintenance across pipe networks. Beyond traditional utilities, space data also supports water-sensitive urban design, such as the development of sponge cities, by guiding blue-green infrastructure placement to strengthen flood resilience. Scaling these applications is key to maximising their impact.

[14] ESPI, Where Sensing Makes Sense, 2026. (Link)
[15] GeoField, Capturing Resilience: How Rainwater Harvesting Shapes Adaptation in African Drylands, 2025. (Link)

- *Green water availability* has low policy appetite, yet is widely addressed by space projects, mostly focusing on agricultural soil moisture and forest evapotranspiration. Consulted experts and literature underscored the **limited prioritisation and underestimation of this use case across water policies, calling for a greater focus and integration into broader water-specific discussions and conferences**. By contrast, ESPI's *Where Sensing Makes Sense* report[16] highlights strong policy emphasis on soil quality, including soil moisture. This disparity between agricultural and freshwater policies may stem from the framing of green water as a human resource rather than a key component of the water cycle, perhaps indicating the need to increase recognition of its ecological importance beyond its direct relationship with human use.
- *Water withdrawals & use for energy infrastructure* – **Despite strong national prioritisation (34% of countries) and the demonstrated potential of space applications, such as for hydropower production and power plant cooling, the uptake of advanced space solutions remains limited in this area**. While this disparity is partly driven by low project maturity (e.g. early-stage pilots and feasibility studies), the absolute number of projects remains low relative to policy objectives. ESPI's *Where Sensing Makes Sense* report highlights a similar disconnect across the broader energy sector, where space project implementation is limited, despite high policy interest.[17] This underscores the need to further leverage space-based solutions for the energy sector in general, with especially significant opportunities at the intersection of energy and water through hydropower production.
- *Groundwater availability monitoring & management* is prioritised by nearly half of the countries (47%) but **remains under-addressed by space-based solutions.** This can be partially explained by the **indirect nature of remote sensing methods for groundwater measurement**. Nonetheless, this use case is gathering increasing interest in the space sector, as demonstrated by the joint GRACE (Gravity Recovery and Climate Experiment) and successor GRACE-FO missions by NASA and DLR. These satellites are used to track subtle fluctuations in water masses by measuring changes in Earth's gravitational pull, enabling estimates of global water storage shifts, including sub-surface changes (e.g. aquifers).[18] Such capabilities demonstrate the potential of space-based solutions to further, and more directly, complement existing in-situ methods of groundwater monitoring and management.
- *Water efficiency* – Despite strong policy appetite, **space project implementation for water efficiency remains limited, largely due to the localised and behavioural nature of the core challenges.** For instance, while *reduction of water losses* is prioritised by 41% of countries in water-related policies, there is an emphasis on localised

[16] ESPI, Where Sensing Makes Sense, Section 3, 2026. (Link)
[17] ESPI, Where Sensing Makes Sense, Section 2, p. 17, 2026. (Link)
[18] NASA, Tracking Groundwater Changes Around the World Using Satellite Gravity, 2025. (Link)

infrastructure (e.g. buildings and pipes), prioritising operational and in-situ methods over space-based applications. Similarly, in the context of *responsible individual water consumption,* solutions are generally more oriented towards promotion, education, and public awareness campaigns. While the contribution of space technology is more indirect in this area, **it holds untapped potential for new forms of public engagement**, such as the use of compelling Earth observation imagery to visualise water scarcity and drive behavioural change.

- *Inland flood risk management and drought monitoring & management* - A strong alignment is identified between policies and space-based projects addressing these two primary disasters. Both are highly prioritised by water policies and widely addressed by space projects of various sizes. However, the profusion of space applications for floods monitoring for instance does not always ensure their efficiency in real-time relief efforts or accurate prediction: this aligns with the findings and recommendations of the *Where Sensing Makes Sense* report[19] as well as the insights from stakeholder consultations, which highlighted the prominence of these use cases but emphasised the need **to transition beyond passive monitoring towards proactive and operational solutions.** For instance, integrating satellite data into the workflows of municipalities can help to forecast supply deficits and enhance preventative water management. As mentioned above, hydrometeorological risks should be interpreted more as drivers than standalone use cases, given the interconnectedness with other use cases.

[19] ESPI, Where Sensing Makes Sense, 2026. (Link)

3 CONVERGING CURRENTS: STRATEGIC PATTERNS AT THE SPACE-WATER INTERFACE

Beyond the opportunities identified through the policy and project analysis, an additional layer of analysis incorporating insights from stakeholder consultations was conducted to capture the broader dynamics and emerging patterns at the intersection of space and data-driven water policy implementation. This section explores the key dimensions along which these strategic patterns emerge:



1. CONNECTING WATER GOVERNANCE AND SPACE SOLUTIONS

The interaction between water governance and space ecosystems is shaped by contrasting institutional logics revealing three interrelated trends: (1) different governance scales (decentralised water governance framework vs. centralised scale of space ecosystem); (2) a selective involvement of water governance actors in space solution design despite their prominent role as end-users; and (3) multiple but fragmented pathways for dialogue between the space and water sectors.



2. PRIVATE WATER COMPANIES: COMPLEMENTARY BUT DISTINCT PARTNERS

Private water companies complement projects led by public water governance actors, but follow distinct priorities: private and public actors rarely appear together in the same project consortium. Additionally, private water companies tend to focus on funding targeted and early-stage solutions and seem to help space projects reach an additional layer of private users.



3. EXPANDING THE FINANCING BASE FOR SPACE AND WATER

Water-related space solutions have the potential to draw on a broader financing ecosystem, with private companies and International Financial Institutions (IFIs) financing open-access platforms, investing in digital solutions and helping create future demand for space solutions and services. However, recognising the extent to which space data helps, and where complementary *in situ* data is needed, remains crucial to avoiding misdirected investments.



4. SPACE AS AN ENABLER OF TRANSBOUNDARY WATER COOPERATION

Space is already supporting transboundary water cooperation through shared and independently verifiable data, yet water diplomacy remains largely absent from space policies and is rarely framed as an explicit objective for space solutions. As space data helps reduce information asymmetries and build trust between riparian countries, its potential extends beyond monitoring to actively supporting conflict prevention and resolution, as well as regional water security.

3.1 Connecting Water Governance and Space Solutions

The relationship between water governance and the space ecosystem is influenced by contrasting institutional logics which reveal three interrelated trends: the mismatch between the predominantly local and decentralised nature of water governance and the more centralised scale of the space ecosystem; the limited co-creation of space solutions with water governance actors despite their prominent role as end-users; and the multiple points of connection through which water and space actors currently engage.

The Local Water Sector Meets a Centralised Space Ecosystem

Unlike other sectors such as Energy, **the water domain is highly decentralised**. Implementation largely occurs at the local level, which can limit economies of scale and reduce the financial and technical capacity of water authorities. Governance is further fragmented between **water supply and sanitation**, typically managed by local utilities at the municipal level (this varies significantly across European countries), and **water resource management**, to a large extent organised at river basin level under EU legislation and supported by governmental oversight, which usually includes the relevant national ministries and/or environmental public agencies. Therefore, effective implementation depends on coordination between local, national and European governance bodies, an important consideration to design effective and tailored space solutions addressing water policy priorities.

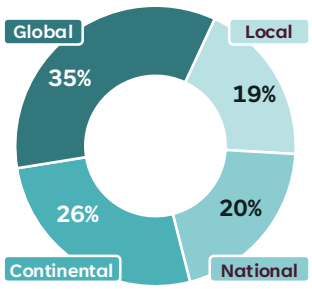


Figure 10 – Scale of implementation of water-related space projects

While water resource management and water supply and sanitation are primarily managed at the local level, the analysed space projects are mainly deployed globally, regionally, or nationally (see Figure 10). 19% of the analysed projects are local in scale, compared to 35% global. However, it is worth noting that, while local-scale projects remain least represented in comparison to the other scales, the share of local-scale projects is significantly higher than in the broader domains analysed using a similar approach in ESPI's previous research, *Where Sensing Makes Sense* [20], where the domain with the highest percentage of local space projects was Climate Resilience with 12%. This reflects the local nature of solutions and the importance of in-situ data in the water domain, with ground verification and the involvement of local stakeholders playing a significant role. Conversely, the prevalence of projects with a geographical scope transcending the local and national level reflects the transboundary nature of water resources (See §3.4).

[20] ESPI, Where Sensing Makes Sense, 2026. (Link)

Limited Co-Creation with Water Governance Actors

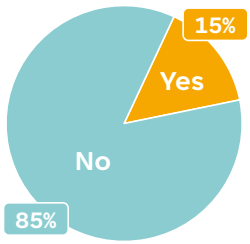


Figure 11 – Proportion of space projects involving water governance actors in their consortium ('Yes')

The analysis points to a limited co-creation of space solutions with their primary end-users. **Only 15% of the analysed space projects involved a water governance actor[21] in their consortium, while many of these same actors constitute nearly half of the primary end-users.** Most consortia are largely dominated by space actors, indicating a potential disconnect between technology providers and the implementation of solutions, therefore posing the risk of limited operational use.[22] Indeed, if space is treated as an external tool rather than an essential component of the solution, it will not be integrated as effectively into the design of solutions for policy-related issues. This points towards the **need for space actors to provide solutions to existing problems emerging at the policy level and involve governance actors in every step of the process.**

There is a **positive sign of involvement of public agencies and municipalities directly in the design of large-scale space projects** that reach a regional or international user base – in Europe, Africa and globally (i.e. multiple countries across multiple continents). Despite involvement in the consortium of just 15% of all projects collected, public agencies and municipalities are still involved in 25% of projects considered as having the highest impact due to their large budget, pluri-annual plan and numerous included activities ('Programmes'). They are also involved in a significant number of smaller scale and early-stage projects such as feasibility studies and pilot initiatives, often backed by large institutional space actors such as ESA.

However, while public agencies and municipalities are the primary end-users of space-powered operational systems, tools or products,[23] they are often not actively part of the team designing them. This poses the risk of a limited uptake and a possible mismatch between needs and available tools. Indeed, water governance actors are involved in almost none of the projects delivering a System or a Platform (7%), which constitute most project types, despite appearing as the primary end-users of Systems and platforms in more than 40% of the cases. Public agencies alone appear as primary users of more than half of them. **Co-creation** has been highlighted as an essential element of water-related space solutions during the consultation and side-event to UN COPUOS in June 2026: **involving water actors**

[21] 'Water governance actors' designate any public local, national, regional and international actors that have a say in water management decision-making or strong influence on the implementation of water management decisions. These include ministries, national/local (public agencies), regions/municipalities, river basin commissions, and others (e.g. public water companies, international organisations). Types of actors and end users are further explained in Annex B.

[22] Further research would be needed to assess whether water governance actors' potential role as contracting authority explains partly their limited involvement in project consortia.

[23] Referred to as "Systems/Platforms" for the purpose of the study, see all definitions in Annex B.

from the start increases buy-in and leads to more impactful solutions.

Additionally, River Basin Commissions stand out as largely absent from the consortia of space projects, only featured in a small percentage of the 250+ projects collected. Stakeholder consultation outputs underscored the essential role of River Basin Commissions as some of the oldest existing International Organisations, who have access to valuable in-situ data, complementary to the data derived from satellite imagery. Most importantly, they transcend borders and act at the level of an entire river basin, without limiting their activities to national borders.[24]

Where Space and Water Actors Connect: Where Does the Dialogue Happen?

The water sector has several points of entry for collaboration involving space technologies and to engage dialogue with space sector actors, which reflects the decentralised nature of the water governance landscape. Dialogue between the water and space sectors occurs at various levels of governance and for different types of activities. This section begins to address the key governance levels and thematic areas identified through the stakeholder consultations.

Through a top-down approach – stemming from policymaking bodies – the water-space dialogue could start with a request for assistance by national ministries to international organisations. The national request for technical support is channelled through the organisation's country offices, when applicable, and transferred to the headquarters, before being assigned to the relevant technical sections of the organisation, including units specialised in geospatial data. **Requests for assistance can also stem from transboundary river basin authorities**, in which case interactions take place directly with the international organisation's relevant section. Through national points of contact, typically at ministry level, space actors can also engage with **international conventions** dedicated to the management of water resources[25] and reach other relevant water organisations through the Convention's network. This proves to be an effective way to collaborate within multi-disciplinary international projects involving several layers of governance and types of institutions.[26]

Conversely, a bottom-up approach can be taken. EO-powered hydrological predictions platforms[27] first engage with disaster management services and hydrometeorological organisations. In this context, data provision is approached as a means to equip local water actors with the tools and knowledge allowing them to develop more sophisticated methodologies and engage with their respective national ministries. The openly accessible

[24] See more on international cooperation on transboundary river basins in §0.
[25] For instance, the Convention on Wetlands, formerly known as the Ramsar Convention. (Link)
[26] As part of the SEEA Ecosystem Accounting initiative (Link) for example, space-water cooperation happens at international and European Union level through Eurostat and the Joint Research Centre of the European Commission (DG JRC).
[27] Such as GEOGLOWS, the initiative of the Group on Earth Observations (GEO) dedicated to global water sustainability. (Link)

hydrological data is then used by various types of actors beyond the water sector, such as mining companies. Additionally, **interactions to make space-derived services themselves evolve according to user needs take place at end-user level**, for instance through Copernicus user forums.

Finally, the dialogue is required for **data verification** purposes, where in-situ data helps address satellite data's technical or practical limitations. Ground verification of satellite data is enabled by local water actors such as NGOs, academic institutions and institutes operating under specific ministries (e.g., agricultural institutes). Beyond verification, **local water actors can further contribute to space projects by providing the land necessary for space actors to conduct the analyses and mapping**. For example, large NGOs can acquire land for conservation and restoration works, which can then be used by space agencies and companies to develop services using the data from the land. In the case of wetland and peatland management, local communities are the primary point of contact for data collection. Peatlands, for instance, can be mapped using in-situ data, while satellite data is valuable for continuously monitoring their health and state of degradation.

3.2 Private Water Companies: Complementary but Distinct Partners

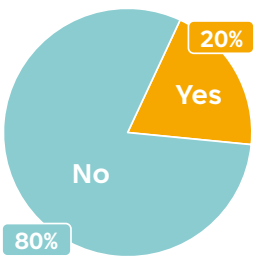


Figure 12 — Proportion of space projects involving private water actors in their consortium ('Yes')

Public and private water actors[28] present a separate set of priorities and are not approached for the same type of projects by space actors. Private water actors participate in 20% of the space projects collected and appear alongside water governance actors in only 5% of all the projects – public and private water actors are rarely both involved in the same projects.

Private water actors appear to mainly fund feasibility studies and pilot initiatives, which tend to be targeted and early-stage projects. They do so usually alongside a public entity. Private water actors are also involved in the consortium of 21% of Programmes, which is considered the highest-impact category of projects. **Similarly to the water governance actors, private water actors are underrepresented in the consortia of the largest part of analysed projects**; they are featured in only 7% of projects resulting in a System or a Platform, with little to no overlap with the projects featuring water governance actors.

The main noticeable difference is that projects involving water governance actors will aim at being primarily used by municipalities and public agencies, while projects involving private water actors are – in addition to municipalities and public agencies – reaching

[28] Private water actor: any commercial actor active in the water sector (e.g., environmental services companies, private water supply companies, consultancies specialised in water management, private water management solutions providers, commercial water services, hydropower companies). This is further explained in the Annex.

another layer of private end-users. These end-users include (non-exhaustively) hydropower producers, private water supply companies, retailers, small-holder farmers, insurers, large producers and cooperatives. Despite this, **municipalities and public agencies still appear (often together) as end-users in roughly half of the projects where a private water actor is part of the consortium**, which still reflects the predominantly public nature of water-related space projects and aligns with one of the main characteristics of the water management sector, which is the prominence of public entities in matters relating to water resources.

3.3 Expanding the Financing Base for Space and Water

While most analysed space projects are financed by public and commercial space actors, stakeholder consultations identified potential avenues to diversify the funding sources of water-related space projects.

3.3.1 International Financial Institutions (IFIs)

International financial institutions (IFIs), such as development banks, invest in digital and innovative solutions supporting the development of water management applications;[29] space-based services can also benefit from these investments in Europe. For instance, developed in coordination with the European Commission's Water Resilience Strategy, the European Investment Bank (EIB) has launched a Water Resilience Programme and a Sustainable Water Advisory Facility (SWAF) expected to mobilise €40 billion of global water investment over the next three years.[30] Climate finance from multilateral climate funds can also reach water-related space projects, provided that the projects benefit low- and middle-income countries.

IFIs also play an important indirect role in creating future demand for space-based platforms. Consulted stakeholders underscore that while investing in open-access platforms for hydrological modelling may generate limited direct revenue for an IFI, it enables the growth of downstream industries which will later require larger-scale financing. Investing now in open-access space solutions therefore expands the IFI's future lending opportunities. Meanwhile, shared space-powered platforms for water resource monitoring or other applications have been made widely available thanks to this initial investment.

3.3.2 Private companies

Regardless of their sector of operation, **private companies** operating in domains such as energy production also have an interest in reducing their water footprint in order to reduce their operating costs. Similarly, they have interest in supporting collective action at river

[29] See for instance, World Bank Group, Earth observation for water resources management: current use and future opportunities for the water sector, 2016. (Link)
[30] EIB's Sustainable Water Advisory Facility (SWAF), launched in March 2026. (Link)

basin level, which includes restoration projects for water ecosystems. If a company depends on water for production, cooling, processing, or as an input (beverage, agriculture, textiles, mining, energy, semiconductors, data centres...), basin degradation is both a consequence of its own operations and a direct threat to their continuity.

Private actors, such as **insurers and large producers,** are also well-positioned to contribute to the design and financing of space solutions. They feature as end-users of water-related space projects, particularly when private water actors (e.g., environmental services companies, private water supply companies) participate in project consortia.

Private companies can also act as sponsors of open-access space-based platforms developed by public entities. By hosting datasets freely on their cloud infrastructure,[31] they help these platforms reaching low- and middle-income countries and communities. According to stakeholder consultations, this approach complements the current reliance of low- and middle-income countries on data generated by Chinese or US providers, often procured through limited three-year access agreements before transitioning to alternative sources. While this model enables cost-effective data access reliant on single funding rounds and avoids the need for long-term data-provision contracts, it may not always guarantee data continuity and comparability over long periods of time. These open-access platforms can help address this gap by providing long-term monitoring free of charge.[32]

Avoiding Misdirected Investments: What Can and Cannot Be Done from Space

Consulted stakeholders underscored the importance of specifying expectations on the type of insights satellite data can provide versus where in-situ data is required, to ensure investments are directed where they can deliver the greatest value. Although an analysis of the technical feasibility of space solutions for the highlighted opportunities is beyond the scope of this study, insights from the stakeholder consultations and further literature[33] allow for a more thorough understanding of where space technologies have demonstrated impact. Specifically, satellite data has been highlighted as valuable to large-scale surface water quantity and quality monitoring (including salination of watershed), forward-looking projection of water availability, water extraction planning, hydroclimatic trends, as well as water infrastructure assessment (e.g. dam stability). However, optical sensors can reach limitations when rivers are narrow or covered by tree canopies, tending to require the integration of space data with other (*in situ)* datasets.

[31] This is the case for GEOGLOWS hydrological datasets, hosted by Amazon as part of the Amazon Web Services (AWS) Open Data Sponsorship Program. (Link)

[32] This observation is also supported by ESPI previous research on Earth Observation data policy.

[33] Examples can be found for instance in: FAO, Leveraging Space Technology for Agricultural Development and Food Security, 2025 (Link); UNOOSA, The Space4water Project: Building Bridges for Knowledge Exchange, 2023 (Link); UNOOSA, Earth Observation for Sustainable Development Goals, 2026 (Link).

3.4 Space as an Enabler of Transboundary Water Cooperation

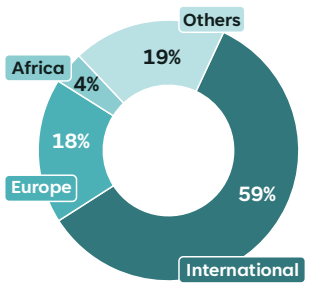


Figure 13 – Main impact regions of water-related space projects

Water is ultimately transboundary in nature and not confined to political boundaries, as reflected in the prevalence of analysed projects with a geographical scope transcending the local and national level. Nearly 60% of the collected space projects have an international impact (i.e. they impact multiple countries one more than one continent, see Figure 13), which is significantly higher than the regional impact in Europe (18%) and Africa (4%).

However, water diplomacy remains a high-level policy concept that has yet to be framed as a concrete objective that space solutions can help achieve, both at policy and implementation levels. Very few space projects specifically address *water diplomacy* use cases – namely *Enhancing international collaboration in business opportunities* and *Cooperation on transboundary water basins* (See Figure 9, §2.2.3). References to *water diplomacy* use cases are also absent from national space policies, despite being prioritised in the water policies and strategies of nearly 30% of countries,[34] some of which even possess an "international water strategy" focused on water diplomacy considerations, such as France and Finland.

Nevertheless, space technologies are already playing an important role in enabling international cooperation on shared water resources. Consulted stakeholders have highlighted that, in many transboundary contexts, negotiations are hindered by either the **absence of data, outdated data, or disagreements over the credibility of datasets.** In this regard, by providing common and independently verifiable information across borders, space-based data facilitate the management of transboundary river basins while also creating the conditions for informed discussions and greater trust between riparian states.

Supporting Transboundary Water Management Across Use Cases

While relatively few space projects collected for the study explicitly address water diplomacy use cases, the analysis reveals a high degree of co-occurrence between *water diplomacy* and other core use cases. Projects addressing water diplomacy often link transboundary cooperation with surface water quality and quantity monitoring, hydrometeorological risk management, and energy infrastructure planning. This area represents a key opportunity for space technologies to transition from monitoring applications towards becoming an active tool for transboundary conflict prevention, joint resource planning, and regional water security objectives.

[34] See Figure 4 and Figure 15

For instance, experts highlight how satellite observations help address the increased governance complexity induced by the transboundary nature of water resources. They explain how satellite-based services have enabled downstream [35] countries across South Asia and Africa to reduce their reliance on upstream countries' data sharing by combining Earth observation with hydrological forecasting. At the same time, upstream countries increasingly use the same openly available datasets, creating a common technical reference for managing transboundary river systems and helping to reduce information asymmetries.

Moving from National Positions to Shared Facts

Stakeholders consistently emphasised the value of Earth observation as a trusted – and to some extent, continuously available[36] – third-party information source, particularly in politically sensitive or conflict-affected basins where access to *in situ* measurements is limited or data sharing is contested. Although disagreements may remain regarding the choice of datasets, open-access satellite observations appear to help shift negotiations towards technical and objective information rather than competing national claims.

In this context, satellite data help break negotiation deadlocks and reinforce water as an entry point for broader cooperation and peacebuilding. For instance, competence centres cooperate with international water basin organisations across Africa and in the Middle East to integrate satellite observations into negotiations over water management processes between riparian countries, while complementing them with in-situ data where necessary (e.g., groundwater data provided by aquifers organisations). Stakeholder consultations also highlighted the importance of governance frameworks: although satellite data can support cooperation everywhere, well-established legal, institutional arrangements and clear funding schemes, facilitate their operational use.[37]

While satellite data alone cannot resolve political disputes, it provides a common foundation upon which trust, cooperation and more effective water diplomacy can be progressively built. The challenge now lies in better aligning space and water sector priorities, for satellite data to deliver benefits throughout the water sector, and beyond.

[35] Here "downstream" and "upstream" should be understood from a river flow perspective, and not a space segment perspective. An upstream country is a country located the nearest to the source of a river.

[36] Recent examples in the US and the EU show the potential limits to the continuous availability of satellite Earth observation data over conflict areas. See the Summary of the 2027 Intelligence Authorization Act, (Link); and the EU delaying release of Copernicus imagery over Gulf of Oman, (Link).

[37] The examples mentioned include the EU's River Basin Management Plans (RBMPs) as part of the Water Framework Directive (WFD) to protect and restore water bodies. (Link)

4 RECOMMENDATIONS

The analysis and stakeholder consultation process demonstrate that the uptake of space technologies in the water sector depends on addressing gaps in governance, collaboration and policy integration. The recommendations below focus on strengthening the connection between space and water sector needs, priorities and capabilities, ensuring that solutions are designed, implemented and scaled in response to operational challenges and through a common agenda.

4.1 Positioning Space as a Core Component of Water Management Solutions

Finding | Awareness of space applications within the water sector is relatively high: nearly 60% of countries analysed refer to space technologies in their water policy documents, while only one fifth of all countries discuss non-space technologies (IoT, AI/ML, drones) without mentioning space. However, this policy awareness does not consistently translate into implementation. Countries managing the largest share of water-related space projects show limited cross-referencing between space and water in their national policies, while others extensively reference both sectors but implement comparatively few projects. This suggests that water and space policies evolve in parallel rather than reinforcing one another: water policies prioritise primarily *water resource management* and *water quality*, while space policies tend to prioritise *hydrometeorological risks*, i.e. established use cases where the value of space is already demonstrated (e.g., droughts and inland flooding).

Recommendation | To encourage the development of solutions that directly address existing water policy challenges and improve the translation of policy priorities into space projects, there is a need to align cross-references between space and water policies around shared priority areas, where relevant. National space policies should explicitly recognise water as a strategic application area, while in countries promoting technology-enabled water solutions without reference to space, strengthening space considerations in water policies can help create entry points for the development of satellite-based solutions aligned by design with sectoral priorities.

4.2 Co-designing Space Solutions with Water Sector Actors

Finding | Active co-creation of space solutions for water applications remains limited: only slightly over 25% of the projects collected include a water sector actor in their consortium, public or private. Public water actors especially, such as municipalities and public agencies, make up nearly half of the primary end-users but remain absent from 85% of the project consortia, risking a disconnect between technology providers and implementation. Projects including private water actors in their consortium not only reach public entities but also an additional set of private end-users, including hydropower producers, water supply companies and small-holder farmers.

Recommendation | Public and private water actors should be systematically involved as co-designers of water-related space projects rather than only consultees. Involving end-users throughout project development would strengthen trust, operational relevance and buy-in of solutions. Successful examples in Europe and Africa already exist, including flood mapping initiatives bringing together space agencies, hydrological services, meteorological agencies and ministries to develop solutions tailored to national and regional needs.

4.3 Connecting Space Solutions with Water Governance Frameworks through Local Actors

Finding | Unlike the space sector, the water sector operates through a decentralised governance structure, with implementation largely taking place at the basin and municipal level. Accordingly, collected local-scale projects, are more prevalent in the water domain than in other sectors, even though nearly 60% of analysed projects operate at the continent or global scale. While global and open-access platforms provide common tools and datasets, local actors often require tailored applications that reflect specific governance arrangements (e.g., river basin authorities) and water management challenges (e.g., wetland health monitoring). Consultations also revealed that stakeholder mapping is rarely funded as part of project design, limiting the understanding of the institutional landscape and the needs space solutions are expected to address.

Recommendation | Space solutions should be designed around the decentralised nature of water governance frameworks, informed by funded and mandatory stakeholder mapping. Systematically identifying governance actors, responsibilities and decision-making processes would allow solutions to be adapted to the water sector policy priorities and operational needs rather than importing external assumptions, therefore increasing their relevance and adoption. To enable this process, the sustained involvement of local and regional water actors – such as country offices of international organisations, international conventions, hydrometeorological organisations, disaster management services, NGOs – is essential. These actors should increasingly act as intermediaries between end users and providers of space solutions to help translate water sector needs into tailored solutions that effectively support decision-making.

4.4 Coordinating Action at the River Basin Level

Finding | River basin organisations remain largely absent from the analysed space projects, despite being uniquely positioned to support national and cross-border water management. The prevalence of space projects with a geographical scope transcending the local and national level reflects the transboundary nature of water and the diversity of actors needed to develop relevant transboundary solutions. As long-standing international institutions, river basin organisations possess valuable governance experience and GIS/HIS datasets, especially in Europe due to the presence of numerous ground monitoring stations. In certain cases, but not systematically, these data are integrated with satellite-based meteorological data mainly for flood and drought forecasting. However, this integration necessitates building the capacities of river basin organisations in a context of often constrained resources, with peer learning as a valuable way to disseminate lessons learned across basins meeting different challenges. Further integration of satellite data with these organisations' ground observations would strengthen the work of river basin organisations beyond flood and drought forecasting, towards freshwater ecosystem management and water quality.

Recommendation | Building on existing networks and peer-learning initiatives between river basin organisations and space actors,[38] communities of practice should evolve to help position river basin organisations as co-designers of space projects for water management. Dedicated, frequent and sustained dialogue should happen between river basin organisations and space solutions providers, with a focus on needs identification and capacity building. Beyond knowledge exchange, this sustained dialogue holds the potential to move from ad hoc collaboration towards shaping a shared agenda between space and river basin management actors, allowing to further integrate space technologies into (transboundary) water basin governance.

4.5 Bridging the Water-Security-Diplomacy Awareness and Operational Gap

Finding | Recent water policies (2019-2025) place growing emphasis on *water security* considerations, which covers *water availability*, *water infrastructure resilience* and *water diplomacy*. While the *water security* cluster remains less prioritised overall, *water infrastructure resilience* is a major policy priority, encompassing *sustainable water supply management, water-sensitive urban design,* and *drinking water production*. Yet, this priority remains under-addressed by space projects. Stakeholders highlighted how the value of space data is often overlooked in water supply and sanitation, traditionally viewed

[38] Such as One Water Vision, an international coalition bringing together the World Meteorological Organisation (WMO) and the International Network for Basin Organisations (INBO). (Link)

as ground-level operations, despite proven applications (e.g., reservoir monitoring, predictive pipe networks maintenance, sponge cities development) that remain to be scaled up.

Meanwhile, nearly 60% of the analysed space projects have an international impact, reflecting the transboundary nature of water resources. Yet, *water diplomacy* remains a high-level policy concept that has yet to be framed as a concrete objective that space solutions can help achieve, either in policy or practice. It is absent from national space policies despite being prioritised in the water policies and strategies of nearly 30% of countries. Few space projects explicitly addressed *water diplomacy*, despite the demonstrated potential and growing role of space data in transboundary basins cooperation, international negotiations and peacebuilding.

Recommendation | The space sector should increasingly develop solutions addressing the operational needs of the water sector, particularly the security and resilience of water infrastructure. This should include stronger engagement municipal level, where water supply is primarily managed, in anticipation of the growing prominence of water security in future policies. In parallel, the role of space technologies in diplomatic efforts and cooperation over shared water resources should be more explicitly recognised and translated into concrete objectives, both in water and space policies and at implementation level.

4.6 Leveraging Private Actors and Finance

Finding | Private water actors (e.g., private water suppliers, water management consultancies) are involved in only 20% of the space projects analysed, with little overlap with public water actors. Yet, in addition to municipalities and public agencies, they provide access to private end-users (e.g., insurers, large producers), some of which are well-positioned to help design and finance space solutions. Stakeholder consultations highlighted that private companies relying on water for their activities also have an interest in reducing their environmental footprint and supporting river basin ecosystem preservation and restoration. Private companies also hold an active role in sponsoring publicly-developed, open-access space-based platforms (e.g., for hydrological modelling), while indirectly stimulating future demand for space solutions.

Recommendation | Opportunities to co-finance water-related space solutions with private companies, and to increase the participation of private water actors in space projects consortia, should be actively explored. Such partnerships would help broaden the reach of space solutions and scale up applications addressing operational needs and preventive action (e.g., leak detection, water supply management).

4.7 Recognising the Limits of Space Technologies and Maximising their Added Value

Finding | The findings highlight the importance of avoiding an overly technology-driven approach to freshwater management. Stakeholder consultations emphasised the risk of "overselling" the potential of the contribution of space technologies to freshwater applications, which can lead to misdirected investment. They consistently underscored how space data, such as EO-derived observations, deliver the greatest value when combined with in-situ data and embedded within broader governance frameworks.

Consultations further underscored that efficiency gains from digital technologies, including space technologies, do not automatically translate into water conservation and, in some cases, can lead to the opposite effect. This reflects the **Jevons Paradox,** which describes how an increase in the utilisation efficiency of a resource (usually caused by technological advancements) creates a reduction in price, subsequently generating a higher demand and usage for that resource (rebound effect).[39] In agriculture, tools enabling more efficient water use often lead to the expansion of the cultivated area instead of reducing overall water consumption. This risk is exacerbated in countries where water allocations remain fixed and low abstraction charges provide little incentive to reduce consumption, with such decisions often entrusted to regional/state governments, in application of the principle of subsidiarity.[40]

Recommendation | Space-based services should be framed as complements to, rather than substitutes for, in-situ data and local expertise. To this end, Earth observation data standardisation efforts should not evolve in complete isolation from in-situ data management practices. Instead, integrating space- and ground-based data will facilitate their mutual adoption by both space and water management communities. At the same time, investments in space-enabled water solutions should be paired with appropriate national policy reforms, notably regarding water allocation frameworks, pricing mechanisms and maximum abstraction limits, to ensure that efficiency gains result in genuine improvements in freshwater management rather than expanded water use.

[39] Mario Giampietro and Kozo Mayumi, Unraveling the Complexity of the Jevons Paradox, Frontiers, 2018. (Link)
[40] European Court of Auditors, Sustainable water use in agriculture: CAP funds more likely to promote greater rather than more efficient water use, 2021. (Link)

4.8 The way forward

The outcomes of this study aim to **inform ongoing and future policy processes at the design, review and implementation stages**. While this study expanded its geographical scope beyond EU Member States, ongoing EU initiatives provide short- and medium-term opportunities to institutionalise the space-water dialogue, building on the sectors' existing awareness of each other's activities.

The implementation of the **European Water Resilience Strategy**[41] – including its Research and Innovation agenda – offers a natural entry point for integrating Earth observation and other space-based services into water policy development and implementation. Notably, the future **Copernicus Water Hub** has the potential to become an important enabler for space & water priorities alignment, as it will become a common entry point for water stakeholders and connect their needs with existing space capabilities. Its impact could be reinforced through capacity-building,[42] as well as the improvement of interoperability, data sharing and user access through the development of national water data portals.[43]

Beyond Europe, international policy processes, including the **UN Water Conference**, also provide opportunities to strengthen the visibility of space technologies as practical tools for freshwater management and cooperation over shared waters. They hold the potential to evolve into recurring gatherings, supported by structured preparation processes involving relevant stakeholders at the regional level (e.g., municipalities, NGOs, private water actors, space solutions providers) to develop concrete programmes and initiatives, building on existing outcomes of water conferences (e.g., UN Water Conference flagship initiatives). The conferences would then provide an opportunity to assess the advancement of the commitment of countries, both at the policy level and in terms of operational implementation, including the uptake of technological innovations.

NEXT STEPS FOR ESPI

[41] EU Water Resilience Strategy Actions Tracker. (Link)
[42] European Water Academy. (Link)
[43] EU Action Plan on Digitalisation in the Water Sector. (Link)

ESPI will participate in the **2026 UN Water Conference** to bring this study's findings to stakeholders across the water sector. ESPI will also attend the next **UN Climate Change Conference in Antalya (COP 31)**, which priorities spanning the food-water-climate nexus, energy transition and resilient urban infrastructure, further underscore the timeliness of the study.

Additionally, while this study is primarily focused on European countries, ESPI sees the value of potentially extending it to a **global outlook**, including non-European regions such as Central Asia or Africa.

ESPI plans to next turn its attention to the **Energy** sector. This study's findings point to significant untapped potential at the intersection of water and energy, with applications supporting hydropower production and power plant cooling remaining comparatively underdeveloped despite their technical potential. This observation is consistent with the findings of ESPI's *Where Sensing Makes Sense* report, which identified comparatively low alignment between policy priorities and space solutions uptake across the energy sector.[44]

[44] ESPI, Where Sensing Makes Sense, Section 2, 2026. (Link)

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ANNEXES

Annex A – Use cases description

The use cases have been divided depending on their application domain into four large categories ('clusters'), three of which are further divided into sub-sets. The use cases are organised into clusters to facilitate analysis and readability and should not be interpreted as scientific classifications. Rather, each use case represents a policy objective that can translate into one or more applications.

Water resource management

This category designates the policy objective of ensuring the sustainable, efficient, and resilient use of water resources across all major socio-economic sectors. This objective encompasses the integrated planning and governance of water in agricultural, urban, and industrial systems, alongside cross-cutting efforts to improve overall water efficiency. It contains the following sub-sets, and individual use cases:

Agricultural water management	The planning, optimisation, and control of water resources within agricultural systems to enhance productivity, resilience, and sustainability
• **Irrigation management** (incl. modernisation of irrigation networks, precision farming, wastewater reuse, rainwater harvesting & reuse)	
• **Aquaculture & livestock water management** (i.e. drinking water for animals, bodies of water for fish)	
Urban water management	The planning, operation, and optimisation of water systems in urban environments to ensure reliable supply, effective wastewater and rainwater management, and sustainable use of water resources.
• **Urban wastewater collection and treatment**	
• **Urban water distribution networks management**	
• **Water-related inland mobility management** (incl. water course & urban waterways navigation)	
• **Urban rainwater collection and reuse**	
Industrial Water Management	The planning, allocation, and control of water use across industrial and energy production processes.
• **Water withdrawals & use for energy infrastructure** (incl. cooling water for power plants and hydropower generation)	
• **Water withdrawals for other industrial use** (incl. textile, food & beverage, cloud computing industry, etc.)	

Water efficiency	The measures and practices aimed at minimising water waste and optimising consumption across users and systems.
• **Promotion of responsible individual water consumption** (incl. behavioural patterns) • **Reduction of water losses** (incl. leakages, through maintenance of existing infrastructure)	

Table 3 – Water Resource Management use cases

Water quality

This category designates the policy objective of safeguarding ecosystem integrity and public health by ensuring that water remains safe, clean, and fit for environmental and societal uses. The objective encompasses the monitoring, prevention, and mitigation of pollution in freshwater bodies and groundwater, as well as the management of sanitation-related risks that affect human health. It contains the following sub-sets, and individual use cases:

Surface water quality	The monitoring, assessment, and management of the physical, chemical, and ecological conditions of rivers, lakes, and other freshwater bodies.
• **Freshwater ecosystem and biodiversity management** (incl. wetland ecosystem, shifts in freshwater species range, freshwater phenology, aquatic, hydro-morphological conditions of water streams) • **Surface water temperature monitoring** (incl. variations caused by thermal pollution from industry practices) • **Surface water pollution management** (i.e. human-induced pollution incl. water course discharges, eutrophication – nitrates/nitrogen and phosphorus pollution - affecting aquatic plants, fish, aquatic invertebrates, algae, as well as pollution caused by natural disasters pollution e.g., flooding)	
Groundwater quality	The monitoring, protection, and management of the chemical integrity of subsurface water resources.
• **Groundwater pollution management** (i.e. human-induced pollution incl. eutrophication – nitrates & pesticides pollution) • **Sea water intrusion risk management** (incl. salinisation from coastal flooding)	
Sanitation	The management of water-related health risks arising from stagnant water, aiming at reducing people's exposure to unsafe water environments.
• **Water-borne disease and pest outbreaks** (incl. stagnant water causing increased mosquito-borne disease & cholera)	

Table 4 – Water Quality use cases

Water security

This category designates the policy objective of ensuring the reliable, resilient & cooperative management of water resources & infrastructures to reinforce long-term stability & shared benefits from water resources. This objective encompasses the safeguarding of water supply through effective management of surface & groundwater resources, the strengthening of water infrastructure resilience to ensure continuity of services, and the promotion of transboundary cooperation & conflict prevention through water diplomacy. It contains the following sub-sets, and individual use cases:

Water diplomacy (or hydro-diplomacy)	The use of water resources as a platform for fostering cooperation, conflict prevention, and shared economic opportunities across borders.
• **Cooperation on transboundary water basins** • **Enhancing international collaboration in business opportunities** (incl. water industry exports)	
Water infrastructure resilience	The planning, design, and management of water systems to ensure the adaptability, durability, and continuity of water services in the face of stressors such as climate variability and urban growth.
• **Sustainable water supply management** (incl. water transport capacity) • **Water sensitive urban design** (incl. sponge cities through flood drainage & retention systems) • **Drinking water production** (incl. drinking water catchment infrastructure protected from pollution, rainwater drainage)	
Water availability	The monitoring, assessment, and management of surface and groundwater resources to ensure sufficient supply across uses and time.
• **Surface water availability monitoring & management** (incl. water basin levels & alteration of flow regimes) • **Groundwater availability monitoring & management** (incl. groundwater & aquifers level) • **Green Water Availability** (incl. soil water storage management, i.e. soil moisture depletion, waterlogging & water-related forest management, evapotranspiration, peatlands)	

Table 5 – Water Security use cases

Hydrometeorological Risk Monitoring and Management

This clusters designates the policy objective of ensuring the observation, forecasting, and mitigation of water-related risks driven by climatic and hydrological variability, aimed at improving preparedness, reducing vulnerability, and enhancing resilience to water-related natural hazards across environmental and socio-economic systems.

Disclaimer: Since these applications act more as underlying drivers than standalone use cases, they should not be evaluated in isolation. Their high prevalence reflects how deeply interlinked they are with other use cases (e.g. drought directly reduces water availability).

This category contains the following use cases:

Cryosphere changes monitoring (incl. glaciers, snow and permafrost melting)
Extreme precipitations patterns monitoring & management (incl. heavy precipitations and variations in precipitations incl. dry spells)
Drought monitoring & management
Inland flood risk management
Management of water-related geohazards (e.g. Landslide, sinkholes, avalanches, mudflows)
Water erosion monitoring & management (i.e., soil erosion due to water movement. Incl. stormwater run-off, sediment load)

Table 6 – Hydrometeorological Risk Monitoring and Management use cases

Annex B – Detailed Methodology

Water and Space Policy Analysis

The document selection follows a structured and multi-level approach to ensure both relevance and comparability across sources and countries. To have a comprehensive understanding of the way freshwater is addressed at policy level, **freshwater use cases** were identified based on the **qualitative screening** of the following documents:

- Reports from the **Intergovernmental Panel on Climate Change** (IPCC AR6 WGII Full Report; IPCC AR6 WGIII Full Report; IPCC AR6 SYR Longer Report) relevant to the topic of water;
- National Strategic Plans on the **Common Agricultural policy (CAP)** from selected EU Countries, chosen for their weight in the total agricultural production and export in the Union and geographically distributed across the continent, with a focus on section 2.1SO5;
- Sections on vulnerability, climate impacts and adaptation measures from **National Communications (NCs)** under the UN Framework Convention on Climate Change;
- **National Adaptation Plans (NAPs)** on climate change;
- For each country **national water policies and regulations** were also selected. Priority was given to official and up-to-date **policy documents** (incl. strategies, actions plans, implementation reports) with a clear relevance to freshwater. The selection also included at least one **nation-wide water-related law or regulation** per country. The national policies were excluded if they were outdated (older than 2016, unless it is the only existing water-specific nation-wide policy), out of scope (e.g., about marine environment) or not a nation-wide policy document (e.g., one-off project or service utility regulation);

The **national water policies** were then analysed to identify **policy priorities** in the field of freshwater, later used in the opportunity and gap analysis. When national water policies identified concrete measures to address a specific use case, this use case was then counted as a policy priority. All non-English documents have been translated in English using the platform DeepL, to avoid differences in interpretation.

Quantitative analysis

In order to assess the level of mutual integration in national space and water policies, a **quantitative analysis** was conducted on a total of nearly 100 policy documents by running, on the one hand, a dictionary of space-related words as well as a dictionary of non-space technologies words on water policies, and on the other hand a dictionary of water-related words on space policies. The results from the automated screening have been checked manually to eliminate irrelevant or double mentions.

The **water policies** have been identified and differentiated depending on their scope into 1) Water policy/strategy, or 2) National river basin management plan. Whenever a non-space technology mention was identified, the corresponding water use case – if specified – was also noted. The level of space mentions is defined as follows: 1-4 mentions (low), 5-10 mentions (medium), 11-23 mentions (high). For non-space mentions: 1 mention (low), 2-3 mentions (medium), 4-6 mentions (high).

The **space policies** encompass national policies in which main focus is space. Defence, Security or Cybersecurity policies mentioning space as a secondary topic have been excluded. When space policies mention "water" without precising a specific use case, no specific target areas were identified. The level of water mentions is defined as follows: 1-2 mentions (low), 3-10 mentions (medium), 10-24 mentions (high).

It is worth mentioning that not all countries in the scope possess a space policy to be analysed: 15% lack a policy which primary focus is space.

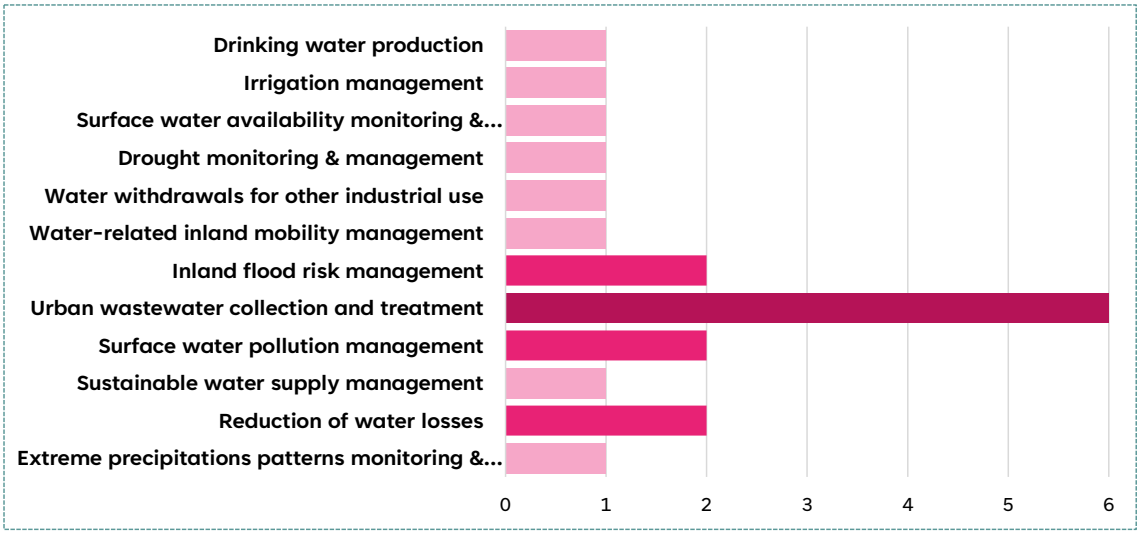


Figure 14 – Use cases referenced by Non-Space Technology mentions in Water Policies

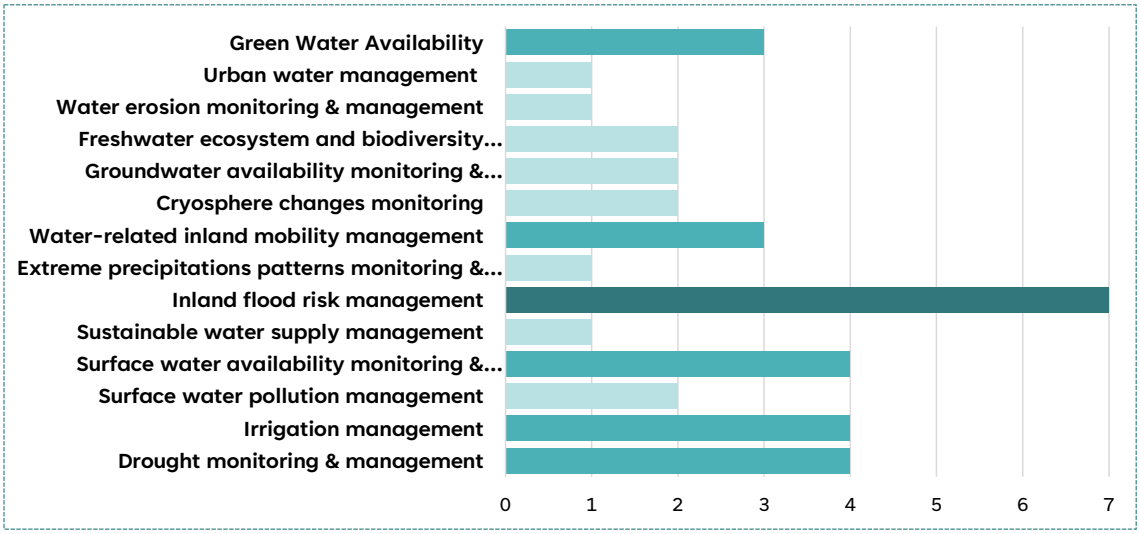


Figure 15 – Use cases referenced by Water mentions in Space Policies

National trends

The countries with the highest number of use cases prioritised in their water and space policies and strategies are **Germany, Croatia, Cyprus and Czechia**, all of which reference at least 50% of identified use cases.

On the other hand, **Lithuania and Estonia** have the lowest number of use cases prioritised at respectively only 17% and 10% of the total.

Space Projects Analysis

To ensure broad and systematic coverage, the mapping of space projects combined **multiple complementary sources**, including European and national project and funding databases, space programme repositories, industry portfolios, application and user-case collections, and relevant impact studies. These were complemented by targeted country-, organisation- and keyword-based searches of national, institutional and water-specific repositories. It should be noted that the comprehensiveness of project collection is limited by the accessibility of project information online, and that the definition of "project" adopted in this report focuses on downstream solutions. Only projects that are **active or planned** were collected. Closed or completed projects were excluded to ensure the database remained up to date.

The initial phase of the project collection was aided by the repository of projects developed for the ESPI Report **"Where sensing makes sense – opportunities for advancing Europe's priorities through Earth observation - ESPI"**. That repository had a wider scope regarding Earth Observation projects for multiple domains different from freshwater; however, some

of the selected projects proved to be relevant also for this report. Therefore, the repository developed for this report build upon some of the findings made before, but expands significantly on the freshwater-specific domain.

The study covers **32 countries**, selected on the basis of their membership in – or affiliation with – the **European Space Agency (ESA)**, the **European Union (EU)**, and/or the **European Economic Area (EEA)**. The countries included in the scope of the study are:

Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, and the United Kingdom.

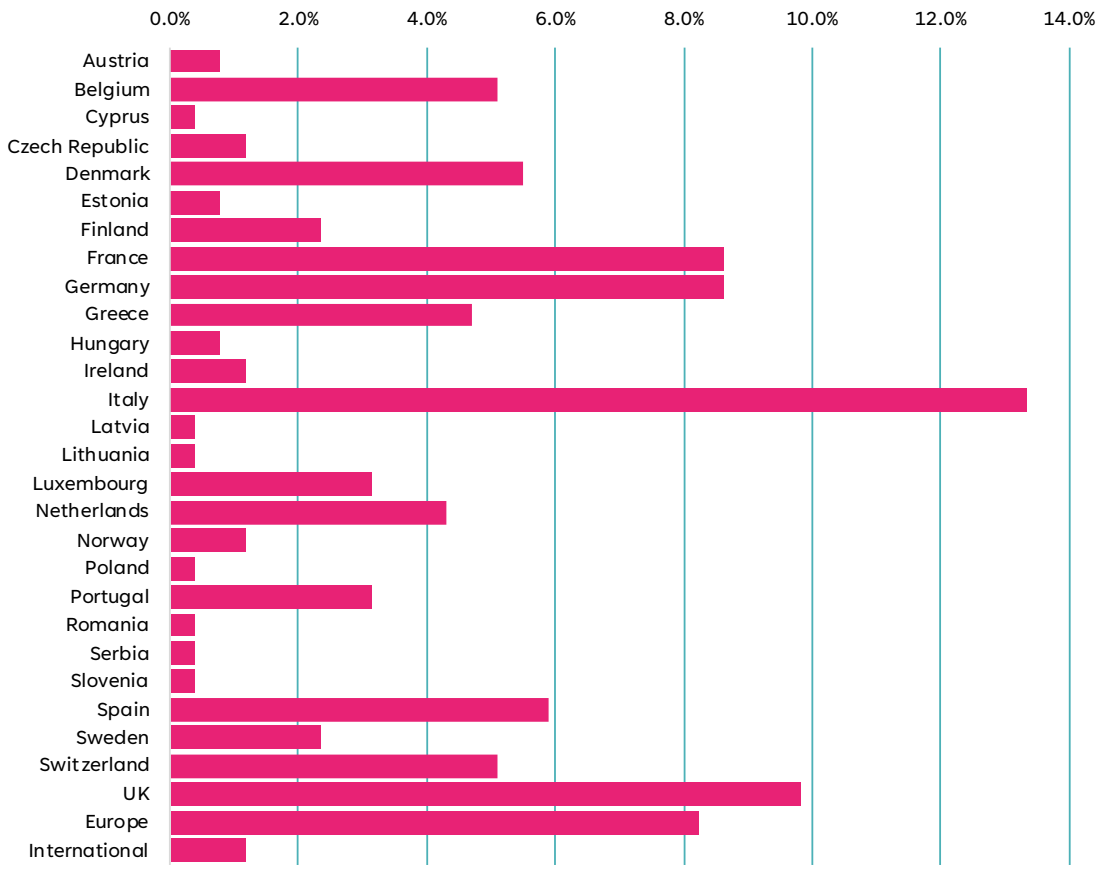


Figure 16 — Distribution of Projects across Managing Regions

Managing Region

The managing region is the region where the prime contractor of the project is based, including for commercial actors. Projects with prime contractors or managing entities outside of the aforementioned list of countries were not included in the database.

Impact Region

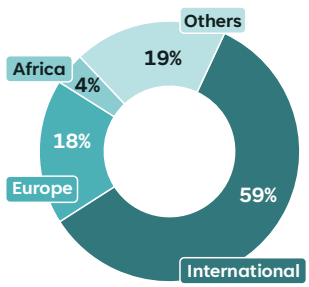


Figure 17 – Main Impact Regions

The area over which the selected projects are being used or implemented is the **Impact Region**. As shown in Figure 17, the most common impact region is international, followed by Others (the combined percentage of individual countries), Europe, and Africa.

One Country (e.g. France, Norway, Spain...)

One Continent (Europe, Africa, Asia, Oceania, South America) (more than one country but all based in the same continent)

International (more than one country but not in the same continent; worldwide solutions; region of use is not mentioned)

Project Scale

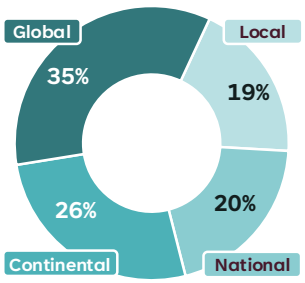


Figure 18 – Scale of projects

The **project scale** is the scale at which the project is implemented. 35% of analysed projects are global in scale, 26% continental, 20% national, and 19% local. Projects with a scope beyond the national and local scale were found to be most common.

Local (project covers a localised region in a country or countries)

National (project covers only one entire country)

Continental (project covers multiple countries in the same continent)

Global (Project covers countries in multiple continents; worldwide solutions; region of use is not mentioned)

Funding Type

The **funding type** is based on the project's funding source(s), as described below. 58% of projects have multiple sources of funding (dual). Many projects are jointly funded by commercial and civil institutional entities (e.g. space agency/European Commission), underscoring the importance of public-private partnerships at the intersection of space and water. 21% of projects are funded by commercial entities alone and 19% are funded by civil institutional entities alone, reflecting a relatively even split

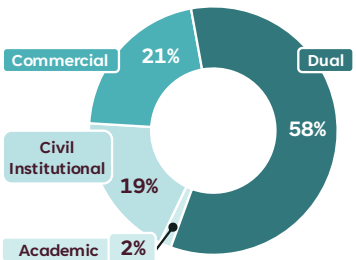


Figure 19 – Funding type

between private and public funding. Finally, just 2% of projects are purely funded by academic institutions.

Academic (fully funded by academic or research institutes)

Civil Institutional (fully funded by governments or part of the "Research and Innovation" pathway of Horizon EU)

Commercial (fully funded by commercial entities)

Dual (projects funded from multiple sources)

Type of Actors

After identifying the entities participating in the space projects consortia (e.g., universities, companies, space agencies, International Organisations), non-space stakeholders specifically active in the water sector were highlighted separately, in order to assess the level of co-creation of space solutions between space and water actors. These actors were divided into two types: the 'water governance actors' and the 'private water actors'.

- **Water governance actors** refer to any public local, national, regional and international actors that have a say in water management decision-making or strong influence on the implementation of water management decisions. These include Ministries and governments, national or local public agencies, regions or municipalities, river basin commissions, and others, including public water companies or international organisations.
- **Private water actors** include any commercial water sector actor (e.g. environmental services companies, private water supply companies, consultancies specialised in water management, private water management solutions providers, commercial water services, and hydropower companies).

Primary End-Users

To understand who the project is designed to serve, the primary end-users for each project were identified and categorised depending on their type of activities, regardless of their thematic focus (here, end-users specifically active in the water sector were not set apart from others). The types of end-users include:

- International Organisations (e.g. UN bodies and international NGOs)
- Civil society actors (e.g. local NGOs and other organisations)
- EU bodies (e.g. European agencies, Directorate-Generals)
- Ministries
- National and local (public) agencies
- Regions and municipalities

- Publicly owned water companies (water service providers)
- Private (i.e. private water companies, insurance companies, energy companies, agribusinesses, farmers)
- Research (e.g. academia, research institutes, think tanks)
- Others (e.g. pipeline operators, water distribution operators)

Level of Implementation

The level of implementation assesses the extent to which space projects currently address policy priorities across identified use cases. This extends beyond the absolute number of projects addressing a given priority to also account for each project's size, scale, and potential for long-term impact.

To define the level of implementation, an **"impact magnitude" score** is determined for each project, placing greater weight on initiatives with strong institutional backing and sustainable funding over small-scale projects with limited funding, though these can be highly impactful at the local level. These individual scores are then aggregated to establish an **overall level of implementation (on a three-tier scale of Low-Medium-High)** for the gap analysis to identify how well policy priorities align with space project implementation. It should be noted that this weighting method was only used for the gap analysis and does not influence the assessment of other project characteristics, such as scale of implementation, impact region, or funding type.

To determine the impact magnitude of a space-based project, a **branching (decision tree) approach** is taken with a scoring system of 1 (lowest) to 5 (highest). A score of 5 is not simply five times a score of 1; the steps are deliberately non-linear. The assessment begins with a categorisation of the "output type". A maximum score is defined for each output type according to the highest potential impact magnitude that each output type can achieve.

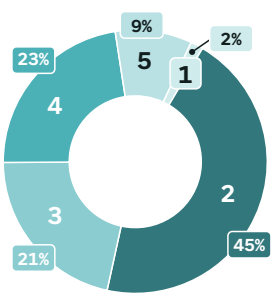


Figure 20 – Distribution of projects based on their impact magnitude score

Across all analysed projects, a score of **2** was the most common impact magnitude (45%), representing early-stage initiatives across all output types excluding programmes. Projects scoring **4** made up 23% of all projects and consisted entirely of systems/platforms, many of which are commercial platforms with institutional consortia stakeholders. Projects receiving a **3** (21%) were represented by pilot initiatives and commercial systems/platforms. Scores at the two extremes were less common, with 9% scoring **5**, and just 2% scoring **1**. All programmes received a score of 5 due to their large budgets and involvement of institutional stakeholders. Conversely, level 1 projects were much smaller in scale, with limited funding, smaller consortia, and purely research-oriented objectives.

The section below explains further the terminology used and the methodology of the Impact Magnitude Score.

The project's status of advancement and scope gives the **Output Type.**

Research/Feasibility: Output is purely theoretical knowledge, or the project is in its initial feasibility assessment phase

Pilot Initiative: Projects in the validating and testing stage but haven't reached full operational capability

System/Platform: Deployed operational system, tool or product

Programme: Wide initiative that involves several activities and long-term objectives. These usually include capacity-building, training programmes, outreach initiatives or a combination of multiple services

The most common output type is **System/Platform**, representing 43% of projects. Half of the projects categorised as System/Platform are funded by commercial entities alone, generally taking the form of an operational tool or value-added service. A third of System/Platform projects are funded through a combination of public and private, with only 15% purely civil institutional funding, thus highlighting the prevalence of the commercial actor for this output type. 27% of projects are research/feasibility studies, which tend to be projects in the earlier stages of development or more theoretical in scope. This output type primarily relies on funding from more than one stakeholder type (dual), often represented by a university collaborating with a governmental actor or a space agency collaborating with a private entity. This category is followed by 21% of projects classified as pilot initiatives and just 9% as programmes, which tend to be larger in scope, involve a variety of stakeholders, and require more funding. Both of these output types have mostly dual funding mechanisms, followed by civil institutional, but none of the collected projects with these output types were found to be funded solely by commercial entities.

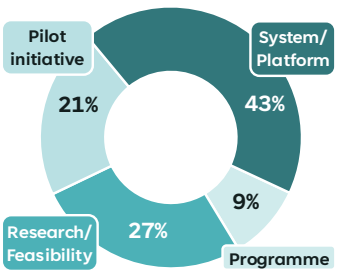


Figure 21 – Output types

The **maximum score** that any type of project can achieve is the following:

- Programme – max score: 5/5
- System/Platform – max score: 4/5
- Pilot Initiative – max score: 3/5
- Research/Feasibility – max score: 2/5

Programme is categorised as the output type with the highest potential impact (5/5) due to the generally large nature of programmes, typically consisting of various different

components (incl. system/platform, research paper, capacity-building initiatives etc) and multi-stakeholder consortia. More weight is given to tangible products with direct outcomes, hence the higher score of **system/platform** in comparison to a research paper, where the outcome of the former provides immediate benefits, whereas the outcome of the latter often requires additional work to derive benefits through the implementation of the findings.

Projects are then assessed against the criteria below. The weight given to each criterion and order of appearance is dependent on the output type of a given project.

Criteria

- **Budget size** – Budget tends to be the primary factor influencing a project's reach and success. It is therefore considered the top criterion in this methodology.
    - This is a **fast-track criterion** for all types, with a specific threshold defined for each. If the budget of a project falls above the specified threshold, the project automatically receives the maximum score for that output type without needing to pass through any additional criteria.
- **Own website** – Whether a project has an established website is often an indicator of its scale and budget.
- **Stakeholders Involved** – This refers to the type of entities involved in the consortium. In this assessment, **Institutional/High-Level Stakeholders** (e.g. Ministries, government entities, big public agencies, space agencies, large international organisations, UN agencies, big companies) hold the most weight. Private actors are divided into big companies (e.g. ICEYE) and small startups. Small startups, small organisations/institutes, and academia hold the same weight as each other, which is assessed as lower than that of the aforementioned entities.
    - More weight is given to **institutional/high-level stakeholders** (e.g. government entities, UN agencies, space agencies, large companies) because they hold more institutional weight and tend to increase the longevity and large-scale capacity of the project, therefore leading to a higher score than small-scale private actors and academic institutions.
    - When **small UN offices** are involved with clear substantial contribution, they can also fall under the "institutional stakeholder" category, but this is assessed on a case-by-case basis.
- **The type of beneficiaries/end-users** – This refers to the target end-users of the solution. Following a similar logic to the above description, **institutional/high-level stakeholders** (e.g. Ministries, government entities, large international organisations or UN agencies) hold more weight than smaller-scale stakeholders (e.g. startups, academia).

Steps:

1. Identify output type: Programme; System/Platform; Pilot Initiative; Research/Feasibility (Study/Paper)
2. Each project output type has its own dedicated decision tree constituting the various criteria listed above. The projects are passed through each criterion until reaching a final score.
3. Once all projects have been assessed, a check is conducted to ensure projects across different output types with the same score are comparable in terms of impact magnitude. This ensures the soundness and accuracy of the methodological approach taken

Output Type	Fast-track budget	Reachable scores	How lower scores arise
Programme	> €10M	4 – 5	Below €10M / unspecified: 5 if an institutional stakeholder is in the consortium, otherwise 4.
System/Platform	> €500K	1 – 4	Own website then institutional stakeholder split the four levels (4/3/2/1).
Pilot Initiative	> €250K	1 – 3	Below threshold: 2 if it has a website, institutional end-users, or an institutional stakeholder; else 1.
Research/Feasibility	> €75K	1 – 2	Below threshold: 2 if it has a website, institutional end-users, or an institutional stakeholder; else 1.

Table 7 – Decision tree logic

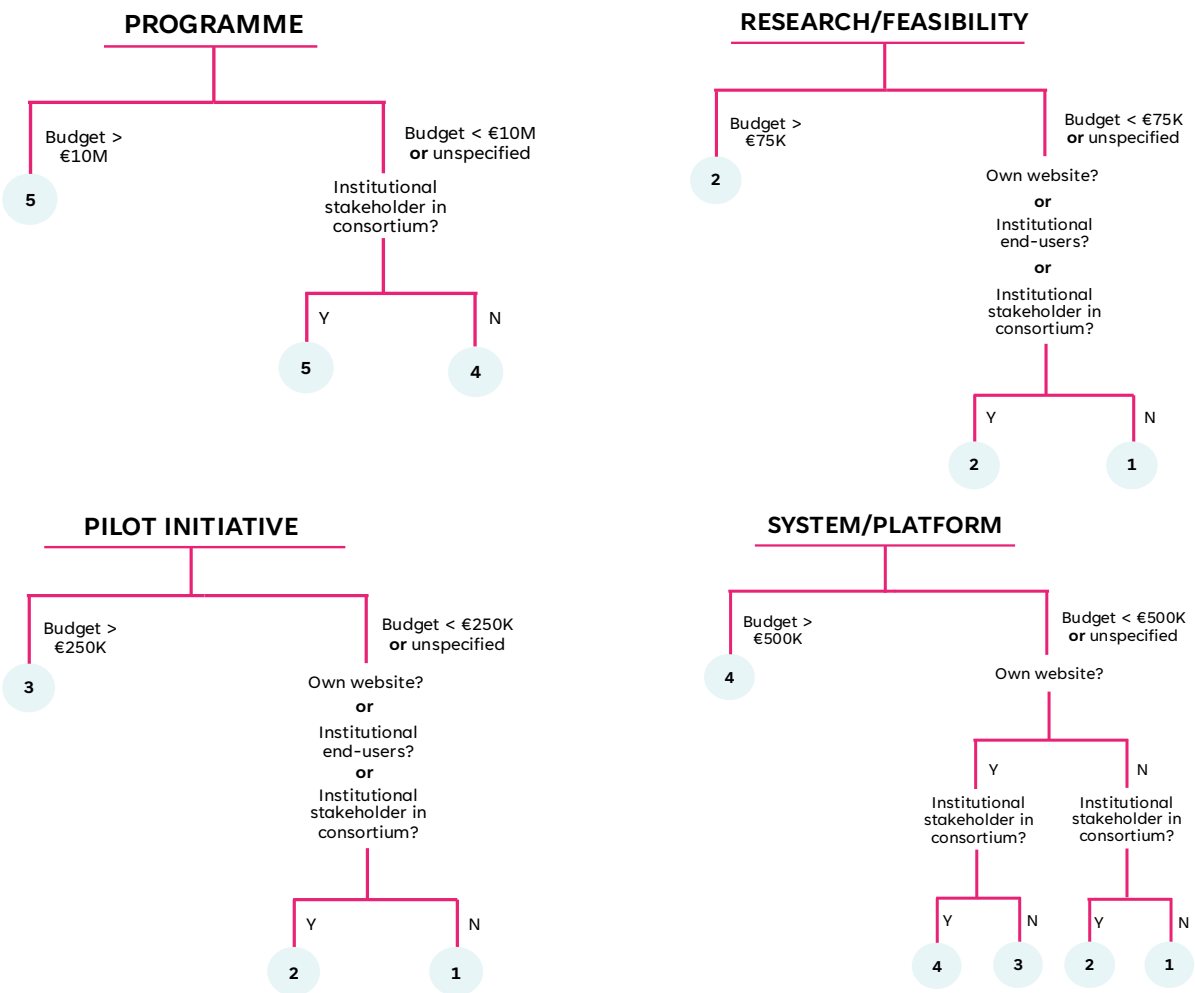


Figure 22 – Decision Trees for Impact Magnitude Scoring

Across all analysed projects, a score of 2 was the most common impact magnitude (45%), representing early-stage initiatives across all output types excluding programmes. Projects scoring 4 made up 23% of all projects and consisted entirely of systems/platforms, many of which are commercial platforms with institutional consortia stakeholders. Projects receiving a 3 (21%) were represented by pilot initiatives and commercial systems/platforms. Scores at the two extremes were less common, with 9% scoring 5, and just 2% scoring 1. All programmes received a score of 5 due to their large budgets and involvement of institutional stakeholders. Conversely, level 1 projects were much smaller in scale, with limited funding, smaller consortia, and purely research-oriented objectives.

Projects with the **same score** were further analysed to test the comparability of their impact and validate the parameters of the decision trees. Two projects for each scoring level were selected as benchmark and compared depending on their financing, scope, type and end-goals. The result of the comparison showed that projects in the same magnitude score have comparable profiles and therefore impact.

Given the results of the comparison above, the weight applied to each project depends solely on the score. The output type does not affect the weight, it only governs which scores are reachable.

The outliers are projects scoring 1; scoring 1 is the result of strong disqualifiers which hinder drastically the impact magnitude of the project. Therefore, the step factor for projects scoring 2 is higher than the step factor for the following scores: the gap in impact between a score 1 and a score 2 is wider than between a score 2 and 3, 3 and 4, and 4 and 5.

Projects scoring 2 are assigned a factor of 5, while projects scoring above 2 are assigned a factor of 3 each, i.e. they each weight 3 times the weight of the previous score. **The end result is that a score 5 outweighs 135 scores 1.**

Score	Step Factor (x prev)	Weight
1	**-**	1,00
2	**5,**	5,00
3	**3,**	15,00
4	**3,**	45,00
5	**3,**	135,00

Table 8 – Project Weighting scoring criteria

Considering a weight formula W(s), where s = score and f = factor;

For $s \geq 2$, $W(s) = W(s-1) \times f_s$

For $s = 1$, $W(1) = 1$

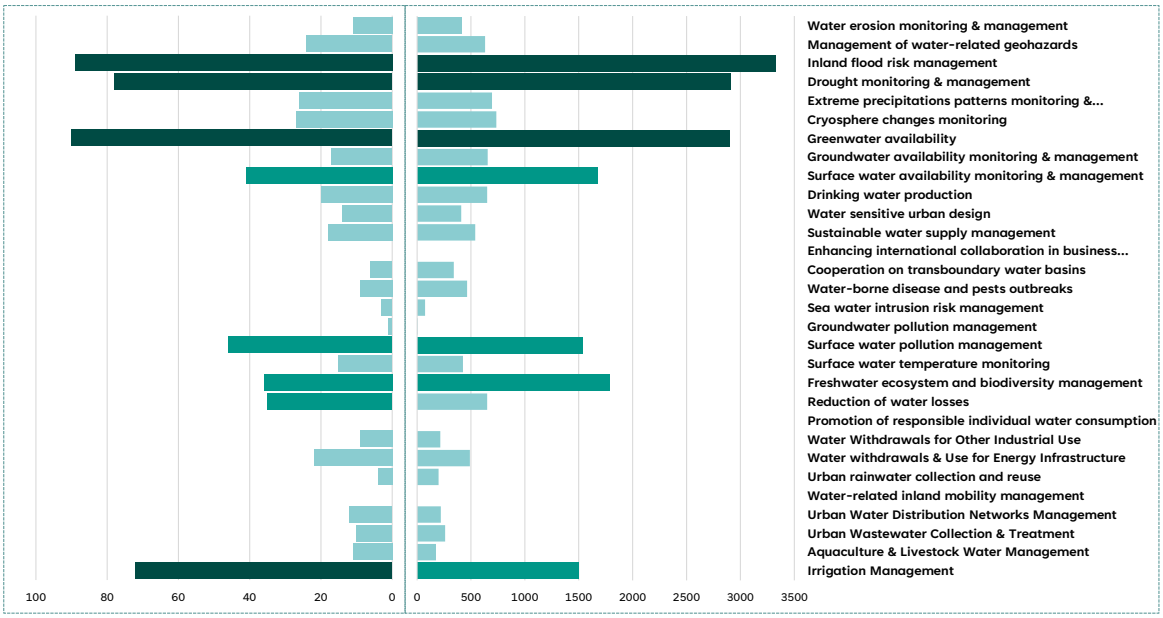


Figure 23 — Project distribution per use case before and after Project Weighting assessment

The **left half of Figure 23** shows the percentage of project distribution per use case in a range from 0% to 100%. This initial assessment is based on the raw number of use cases tackled by each project without taking into consideration the Impact Magnitude of each one.

The **right half of Figure 23** shows the project distribution per use case after the project weighting applying the scoring criteria described in Table 8 (the projects have been categorised using a three-tier scale from low to high and weighted in a range from 1 to 135 depending on each project's "Impact Magnitude" scoring applying the criteria of Table 7).

The Impact Magnitude Scoring method allowed to bring more granularity to the analysis by better determining the actual impact of small projects which ticked many use cases, allowing them to not overflow the assessment (e.g., small-scale agricultural projects which systematically ticked *Green Water Availability* and *Irrigation Management*).

Notably, applying this method changed the order of the top three use cases: (1) *Inland Flood Risk management*, (2) *Drought Monitoring & Management*, (3) *Green Water Availability*.



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